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F29SO – Software Engineering

Stage 2 Report: The Product

PowerHouse – The Smart Home Application

Team Members

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1. INTRODUCTION

1.1 Purpose

This document presents the final stage of development for PowerHouse, a smart home application developed by PowerHouse Inc. to provide seamless control and automation of home environments. Building upon Stage 1, this report details the final product, usability evaluation, marketing strategy, and project assessment.

It highlights the progress made since Stage 1, focusing on:

- The completion of core system features and functionalities.
- Enhancements made to the user interface and experience (UI/UX).
- A thorough usability evaluation with real users.
- Marketing strategies for positioning PowerHouse in the smart home industry.
- Project evaluation, reflecting on challenges, teamwork, and outcomes.

1.2 Scope

The scope of this project is to deliver a fully functional, user-friendly, and robust smart home application, ensuring it meets the functional and non-functional requirements while addressing usability concerns identified in Stage 1.

The key focus areas in this stage include:

- **Finalizing system design & implementation** – Completing development and ensuring stability and efficiency.
- **Testing and usability evaluation** – Conducting a full-scale usability assessment to refine the user experience.
- **Marketing strategy execution** – Developing and presenting **marketing materials** to demonstrate commercial potential.
- **Project evaluation** – Analyzing project management, collaboration, and technological choices.
- **Final product demonstration** – Presenting the core features, system architecture, and usability aspects at the Expo event.

The application now includes:

- **Automation & Scheduling** – Set rules for automated device control.
- **Energy Monitoring** – View and analyze household energy usage.
- **User Profiles** – Distinct roles for super-users and sub-users.

- **Security & Accessibility** – Multi-factor authentication and compliance with accessibility standards.

This stage ensures that PowerHouse is market-ready, incorporating insights from testing and refining it for deployment.

1.3 Overview

PowerHouse is a centralized smart home application that allows users to remotely control, automate, and monitor connected home devices. The final report provides insights into the full software development cycle, from planning and implementation to evaluation and demonstration.

1. **Introduction**
   Outlines the purpose, scope, and goals of the PowerHouse smart home application, highlighting key features and the report's structure.
2. **Final Application Overview**
   Details the product's core functionalities (e.g., energy monitoring, device management) and requirements using the MoSCoW prioritization method.
3. **High-Level Overview**
   Describes the system architecture, interface design, database structure, and graphical interface, including diagrams and technical frameworks.
4. **Marketing Analysis**
   Covers market strategies, target demographics, competitor analysis, SWOT, PESTLE, and the 7Ps marketing mix to position PowerHouse in the smart home industry.
5. **System Testing**
   Explains the testing processes, including unit tests and GitHub CI/CD pipelines, ensuring code quality and security.
6. **Project Evaluation**
   Assesses team organization, implementation approach, tools used, and product robustness, including a requirements fulfillment table.
7. **Usability Study**
   Summarizes user testing methodologies, findings, and improvements based on participant feedback to enhance the application's usability.
8. **User Guide**
   Provides step-by-step instructions for users to navigate and utilize the application's features effectively.

9. **Appendix**

   Includes supplementary materials like meeting diaries, marketing assets, database screenshots, and usability study forms.

10. **References**

    Lists sources cited throughout the report to support research and data

2. Final Application Overview

2.1 Product Scope

The PowerHouse Smart Home Application is a web-based platform that enables users to monitor and control Wi-Fi-enabled smart home devices remotely. The system focuses on energy efficiency, home automation, and user-friendly controls. The final system integrates a robust backend, a responsive frontend, and secure cloud-based storage to ensure seamless operation.

Core Functionalities:

- **Energy Monitoring** – The system provides real-time and historical energy consumption reports for each connected device.
- **Device Management** – Users can add, remove, and organize their Wi-Fi-enabled smart devices within the app.
- **Remote Control** – Users can turn devices on/off, adjust settings, and manage smart home appliances from anywhere.
- **Automation & Scheduling** – Users can set up automated schedules for devices. For example, they can program lights to turn off at night or thermostats to adjust temperature based on the time of day. Additionally, conditional automation is supported (e.g., turning off devices when no motion is detected).
- **Multi-User Access** – The system differentiates between Super-Users (homeowners/admins) and Sub-Users (family members or guests). Super-Users have full control, while Sub-Users can only access assigned devices.

2.2 Requirements

This section covers the final state of the project's user and system requirements using the MoSCoW prioritization method:

- Must
- Should
- Could

i. Functional User Requirements:

1. F-UR-1 Device Management

F-UR-1-1 Device Pairing (During Setup)

- The user must be able to connect their devices to the system network during setup.

F-UR-1-2 Device Pairing (After Setup)

- The user **must** be able to connect their devices by manually entering the devices' IP or MAC addresses (Refer to F-SR-2-1) or by automatic detection of Wi-Fi enabled smart devices.

F-UR-1-3 Device Monitoring

- The user **should** receive alerts on frequent disconnection, low battery levels or significantly degraded performances of any device.

2. F-UR-2 User Management

F-UR-2-1 Home Registration

- The super user **must** be able to register their home to the application with a Google account.
- The super user **must** verify the registered Google Account.

F-UR-2-2 User Login

- Every user **must** log in with the registered Google account by typing a code they receive using a verification mail on the Google Account before they can begin to use the application.

F-UR-2-3 User Hierarchy

- The user who registers the home on the application **must** be assigned the profile of super user.
- Every subsequent user to log in on to the application **must** be assigned the profile of sub-user.

F-UR-2-4 User Profile

- Every user **must** choose the profile allocated to them by the super-user when they log-in onto the application.
- The user **must** be able to view and alter their profile settings.

3. F-UR-3 Device Control

F-UR-3-1 Remote Device Commands

- The user **must** be able to control Wi-Fi enabled smart devices remotely through the application.
    - Example: Turning the smart light on or off.

F-UR-3-2 Real Time Feedback on Device Commands

- The user **should** be able to see real time feedback confirming the commands' success or failure.
    - Example: Connection dropped while trying to edit

F-UR-3-3 Group Device Commands

- The user must be able to control a group of devices at the same time through the application.
    - Example: Turning on all the lights inside the bedroom.

4. F-UR-4 Automation and Scheduling

F-UR-4-1 Command Scheduling

- The user must be able to set basic scheduling and automation commands.
    - Example: The light in the dining room should turn on at 6 PM.

F-UR-4-2 Condition-triggered Actions

- The user should be able to define conditional triggers that prompt device actions. (Refer to F-SR-5-2)
    - Example: If then else and When conditionals.

5. F-UR-5 Energy Consumption and Monitoring

F-UR-5-1 Energy Usage Reports (Super User)

- Super users must be able to view daily, weekly and monthly energy usage reports, per device energy usage reports and overall energy usage reports.
- Super users must be able to compare the trends in their daily, weekly, and monthly energy usage reports.

F-UR-5-2 Energy Usage Reports (Sub User)

- Sub users must be able to view daily, weekly and monthly energy usage reports, per device for each device that they are assigned to by the super user.
- Sub users must be able to compare the trends in the daily, weekly, and monthly energy usage reports for devices assigned to them by the super user and overall energy usage.

F-UR-5-3 Real-time Energy Usage Reports

- Super users should be able to view real time energy consumption from IoT devices, per device for all devices and overall energy consumption.
- Sub users should be able to view real time energy consumption from IoT devices, per device for devices assigned to them by the super user and overall energy consumption for all the devices assigned to them.

F-UR-5-4 Energy Saving Recommendations

- Users should be able to view the devices using the most energy and be provided with recommendations to reduce energy consumption.

F-UR-5-5 Energy Saving Achievements

- Users could be incentivized with Energy Usage Achievements

F-UR-5-6 Energy Saving Goals

- Users should be able to set personalized energy saving goals.

F-UR-5-7 Statistics Sharing on social media

- Users must be able to share their energy consumption reports and energy consumption statistics on social media.

6. F-UR-6 Real Time Energy Consumption Monitoring

F-UR-6-1 Energy Usage Reports (Super User)

- Super users should be able to view daily, weekly and monthly energy usage reports, per device energy usage reports and overall energy usage reports.
- Super users should be able to compare the trends in their daily, weekly and monthly energy usage reports.

F-UR-6-2 Energy Usage Reports (Sub User)

- Sub users should be able to view daily energy usage reports, per device for each device that they are assigned to by the super user.
- Sub users should be able to compare the trends in the daily, weekly and monthly energy usage reports for devices assigned to them by the super user and overall energy usage.
- Sub users should be able to compare the trends in their daily, weekly and monthly energy usage reports.

ii. Non-Functional User Requirements:

1. NF-UR-1 UI/UX

NF-UR-1-1 Dashboard Customization

- The user could have the freedom to rearrange and design the interface elements.

NF-UR-1-2 Themes and Appearance

- The user could be able to choose from a set of preset themes and color palettes offered by the application.

NF-UR-1-3 Intuitive User Interface

- The user should be able to navigate and easily understand the main functions offered by the application.

NF-UR-1-4 Tips and Tutorials

- The user should be offered tips and tutorials on how to operate the app efficiently and properly.

NF-UR-1-5 Accessibility Settings

- The user should be able to alter the features of the application to accommodate their preferences.

2. NF-UR-2 User Profile Security

- The user should be able to set a profile lock with passcodes made by the user itself.

iii. Functional System Requirements

1. F-SR-1 System Network

- The system must support the integration of an existing Wi-Fi network to act as a central hub for device management.

2. F-SR-2 Device Management

F-SR-2-1 Device Pairing (During Setup)

- During setup, devices must connect to the system network using typical Wi-Fi protocols.
- The system must ensure that each device receives a valid IP address or MAC address through DHCP, ensuring proper network communication.

F-SR-2-2 Device Pairing (After Setup)

- The system should allow for two methods of pairing devices:
  - Manual entry of IP or MAC address assigned to the device at setup.
  - Automatic detection of the device and prompting the user to act.

F-SR-2-3 Device Status Monitoring

- The system could track the status of connected devices.
- This includes the monitoring of factors

F-SR-2-4 Device Status Updating

- The system must detect and update device status when faced with key issues such as a device going offline.

F-SR-2-5 Device Status Alerts

- The system should trigger and send alerts to the user for any device that experiences frequent disconnections or degraded performance.

3. F-SR-3 User Management

F-SR-3-1 Home Registration

- The system must allow the super user to register their home to the application with a Google Account.

F-SR-3-2 User Login

- The system should enable users to log in with the registered Google account on the application, followed by entering a verification code received by mail.

F-SR-3-3 User Hierarchy

- The system must assign the role of Super User to the user who registers the house to the application.
- The system must assign the role of Sub User to every subsequent user who logs in to the application.

F-SR-3-4 Device Status Updating

- The system should trigger and send alerts to the user for any device that experiences frequent disconnections or degraded performance.

F-SR-3-5 Email Verification

- The system should verify subsequent user logins for other users through a verification mail.

4. F-SR-4 Device Control

F-SR-4-1 Remote Device Commands

- The system must enable the users to control Wi-Fi enabled smart devices remotely through the application.

F-SR-4-2 Real Time feedback on Device Commands

- The system should provide the user with real time feedback confirming the commands' success or failure.

F-SR-4-3 Group Device Commands

- The System should enable users to group devices together to be controlled at the same instance.

5. F-SR-5 Automation and Scheduling

F-SR-5-1 Command Scheduling

- The system should support scheduling of actions by the users.
- The system should support automation in the following two ways:
    - Implementing custom made rules along with an inbuilt timer for keeping states and times.
    - Connecting to the IFTTT API for easing the automation process, allowing the user to define more complex rules.

F-SR-5-2 Condition Triggered Actions

- The system could enable the user to define conditional triggers that prompt device actions.
- The system could support the following conditional triggers:
    - Basic "if-then" conditionals:
        - The system could allow for basic "if-then" conditionals to define how devices respond to basic specific conditions. These conditions can be based on the status of the connected device.
    - Time based conditionals:
        - The system could allow the user to set conditionals based on time of the day or timers.
    - Device state-based conditionals:
        - The system could allow the user to set conditions based on the state of a given connected device.
    - Combination of conditional triggers:
        - The system could allow the user to nest multiple conditional triggers.

6. F-SR-6 Energy Consumption and Monitoring

F-SR-6-1 Live Energy Consumption Data

- The system should receive live energy consumption information from the devices.

F-SR-6-2 Energy Usage reports

- The system should be able to display real-time, daily, weekly, and monthly energy usage reports, as per the needs and type of user.
- The system should generate reports that are interactive and customizable.
- The system should be able to retrieve and store energy consumption data in JSON format, from a particular day for 30 days before deletion.

F-SR-6-3 Energy Usage Analysis

- The system should be able to analyze the energy consumption data and make conclusive findings on energy usage distribution.
- The system should be able to compare trends in the energy consumption data over time and provide the user with conclusive comparative findings.

F-SR-6-4 Energy Saving Recommendations

- The system should be able to use the above-mentioned conclusive findings on energy usage distribution (Refer to F-SR-6-3) to provide users with recommendations to reduce consumption.

F-SR-6-5 Energy Usage Achievements

- The system could use the above-mentioned conclusive comparative findings (Refer to F-SR-6-3) to notify users about achievements surrounding improving sustainability levels.

F-SR-6-6 Statistics Sharing on Social Media

- The system could be integrated with social media to let users share their energy consumption reports and energy usage achievements.

7. F-SR-7 Notifications

- The system should send the user notifications on Device Monitoring (Refer to F-UR-1-3), Real Time Feedback on Device Commands (Refer to F-UR-3-2), Energy Saving Recommendations (Refer to F-UR-5-4), Energy Usage Achievements (Refer to F-UR-5-5), etc.

iv. Non-Functional System Requirements:

1. NF-SR-1 Accessibility Settings

- The system should provide settings to accommodate preferences such as:
  - Adjusting Text Size
  - Adding Colorblind Filters
  - High Contrast Mode
  - Read-aloud Content

2. NF-SR-2 Dashboard Customization

- The application should allow the user to design interface elements such as tile sorting and rearrangement.

3. NF-SR-3 Themes and Appearance

- The application should allow the user to choose from a set of preset themes and color palettes offered by the application.

4. NF-SR-4 Intuitive User Interface

- The system should be easy to navigate through and its main functions should be easily understood by the user.

- The system should group the main functions together and make it navigable through the home screen.
- The system should ensure all the main functions can be accessed from any page.

5. NF-SR-5 Security and Privacy

- The application should be compliant with data regulation policies based on the regions it is deployed and redistributed in.
- The user data should be encrypted and stored on our database using libraries such as BCrypt.

3. High Level Overview

This section discusses the high-level overview of system architecture.

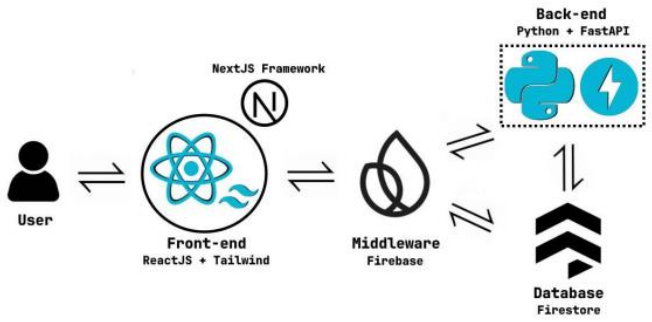


Figure 1: Diagram illustrating the workflow of architecture system.

3.1 Application Architecture

The system architecture outlines the workflow from user interaction to data processing. When a user performs an action, such as logging in or controlling smart devices, the request is handled by the front-end, which is built using ReactJS and Tailwind CSS within a Next.js framework. This request is then sent to Firebase middleware, which acts as an intermediary between the Front-end, Back-end, and Firestore database. Depending on the request, the middleware either retrieves data from the database or triggers a logical process in the back end. If back-end processing is required, relevant data is computed and returned to the middleware, which then delivers the response back to the front-end, completing the user interaction. This architecture ensures efficient and smooth communication between components and efficient request handling.

3.2 Interface Design

The user interface of our application is designed using React.js with the Next.js framework, ensuring a modular, scalable, and maintainable architecture. React's component-based structure allows us to create reusable UI elements, making it easier to manage complex views while ensuring consistency across different parts of the application.

We leverage Material-UI (MUI) and TailwindCSS to create a modern, responsive, and visually appealing design. MUI provides a rich set of pre-styled components that enhance accessibility, usability, and consistency across the application, while TailwindCSS offers flexibility in styling for rapid customization and responsiveness across different devices and screen sizes. This ensures an intuitive and seamless user experience when interacting with smart home devices.

The use of Next.js provides additional performance benefits, including server-side rendering (SSR) and static site generation (SSG), which improve loading speeds and optimize the rendering of real-time data. MUI's built-in theming and Tailwind's utility classes ensure a cohesive and customizable design system, making navigation smooth and user-friendly. Users can efficiently control smart home devices, monitor energy consumption, and receive real-time updates with a well-optimized, performant, and aesthetically refined interface.

3.3 Database Design

Firestore was selected as the primary database for our application due to its seamless integration with firebase services, real-time synchronization, and scalability.

The PowerHouse database design comprises 7 interconnected tables that support profile management, device control, automation schedules, and energy tracking functionalities.

i. Profile Table

The Profile table stores user details such as user ID (primary key), username, password, role (e.g. super-user, sub-user), and allocated devices referencing the Devices Table (foreign key).

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
user_id	INT	Y	N
user_name	string	N	N
user_password	string	N	N
user_role	string	N	N
allocated_devices	array	N	Y

ii. Device Table

The Device table stores smart device information comprising a unique ID (primary key), IP address, and device name.

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
id	INT	Y	N
ip	string	N	N
name	string	N	N

iii. Automation Table

The Automation table manages automation details with fields for automation ID (primary key), linked device IDs as references to the Device table (foreign key), automation name, device status, and trigger conditions.

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
device_ids	array	N	Y
id	INT	Y	N
name	string	N	N
status	string	N	N
triggers	string	N	N

iv. DeviceGroup Table

The DeviceGroup table enables grouping of devices and includes a user-group ID (primary key), group name, status, and foreign key references to both devices and associated user profiles.

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
devices	array	N	Y
user-group_id	string	Y	N
name	string	N	N
status	string	N	N
user	array	N	Y

v. DailyEnergy Table

The DailyEnergy table stores power usage data per day along with timestamps, using a unique ID as the primary key.

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
id	INT	Y	N
power_usage	INT	N	N
timestamp	timestamp	N	N

vi. MonthlyEnergy Table

The MonthlyEnergy table stores power usage data per month along with timestamps, using a unique ID as the primary key.

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
id	INT	Y	N
power_usage	INT	N	N
timestamp	timestamp	N	N

vii. EnergyGoal Table

The EnergyGoal table holds user-defined energy consumption targets, consisting of a unique ID (primary key) and a goal value.

Attribute	Data Type	Primary Key (Y/N)	Foreign Key (Y/N)
id	INT	Y	N
goal_value	INT	N	N

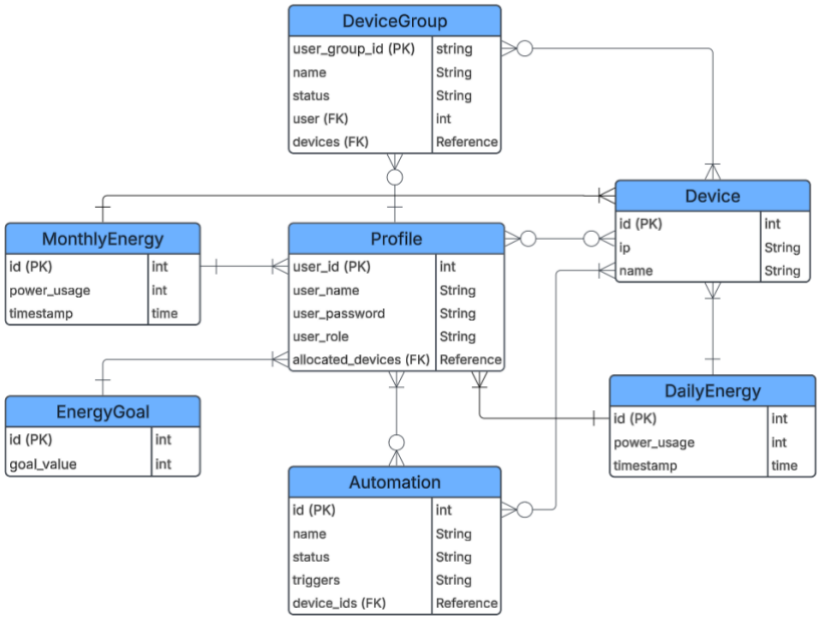


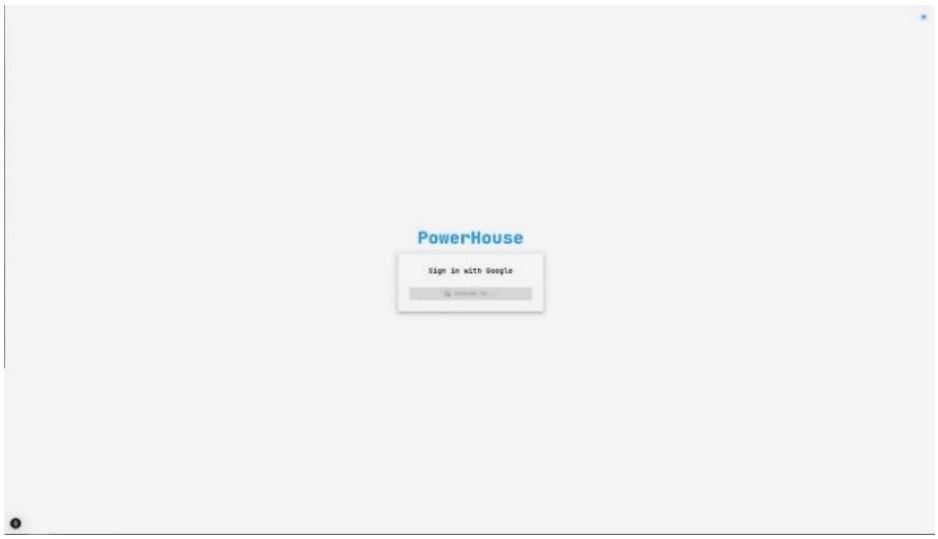
Figure 2: Database diagram illustrating the relationships between tables.

3.4 Graphical Interface Analysis

This part of the report runs over the application's user interface and the functions ready for use.

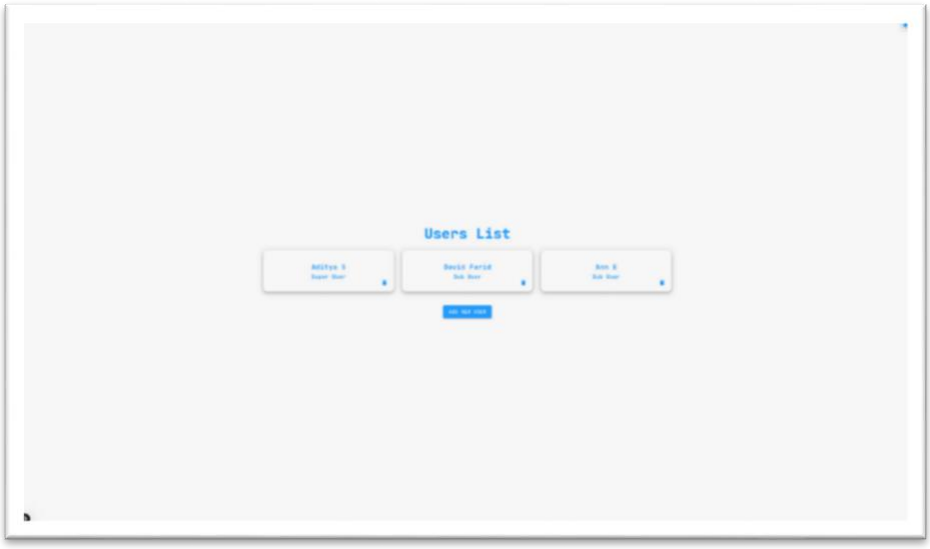
I. Login Screen

Upon application bootup, the user is greeted with a login screen backed by Google's OAuth2 login method. Users must log in via this authentication method before they are able to use the application's functions. Additionally, a light/dark mode switch is offered at the top right of the screen.



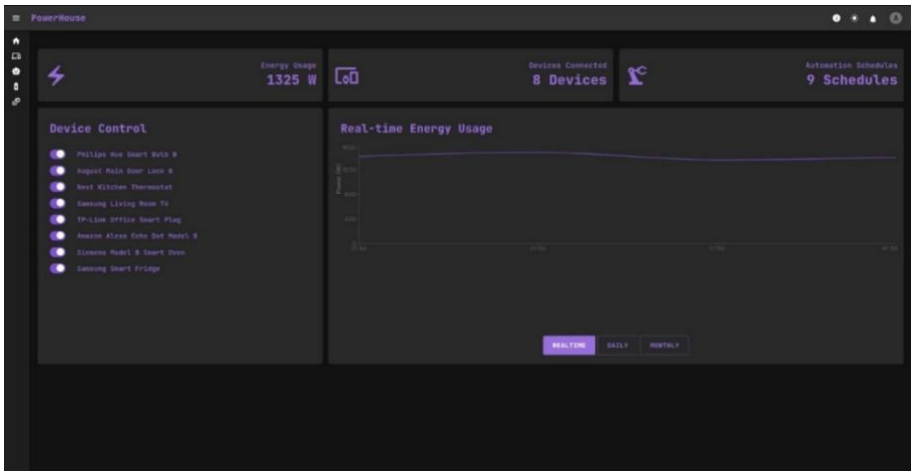
II. Users List

Upon logging into the application using one's Google email address, the user is then prompted to select (or create) a user. The user profile is where the user can use all the functionality the app provides.



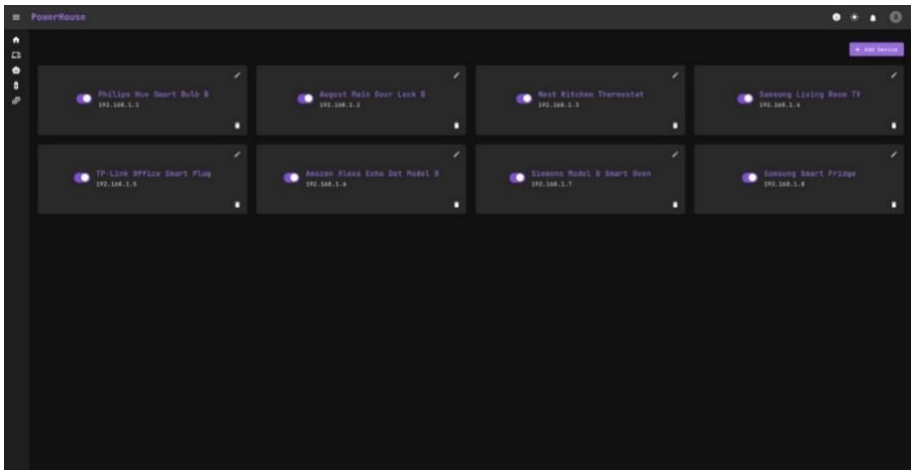
III. Main Dashboard

Once the user has selected a profile, they are directed to the main screen of the application, where device control, real time energy monitoring, and automation scheduling information are displayed. Additionally, at the top right of the screen, quick shortcuts to access the user manual, light/dark mode, notifications tab and user profile are shown. To the left of the screen is a hamburger menu that shows all the application's tabs, and this menu persists throughout the rest of the pages.



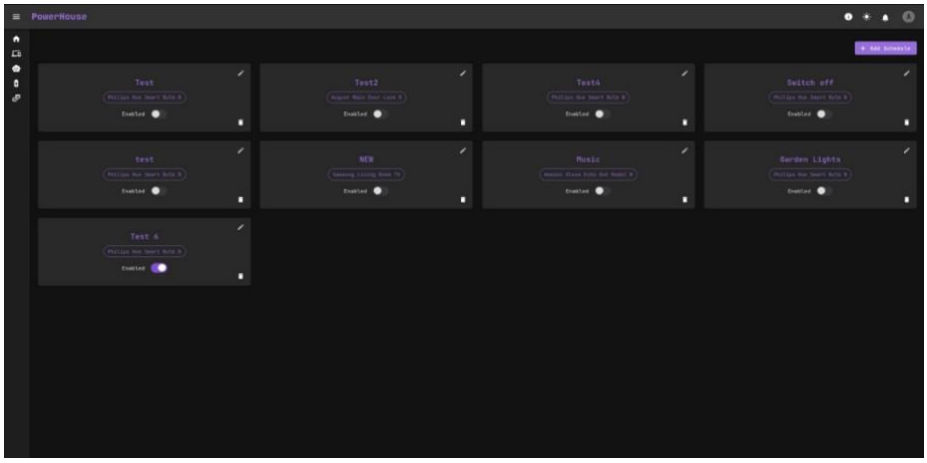
IV. Device List

The second tab is where the connected devices, and their respective data, are displayed. Over here, the user can edit each device according to their needs, toggle status, learn slightly more advanced information about each device (such as its IP address in the home network), and modify the availability of these devices as shown by the trash-can icon. Furthermore, devices can be added using the [+ Add Device] button via Wi-Fi connection, however in this case the devices were already setup via JSON.



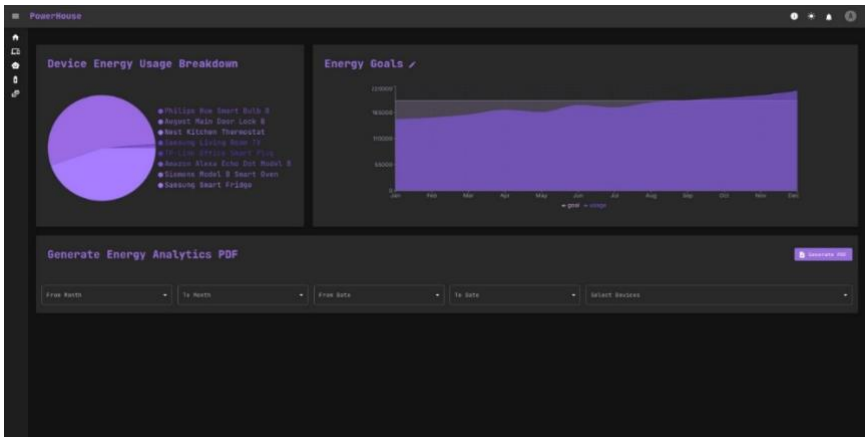
V. Automation Schedules

This part of the application is where automation schedules are displayed and created. Over here, the user can choose to determine a time-based activation schedule for any device specified, binding it to a specific timeframe as well (specific up to the minute).



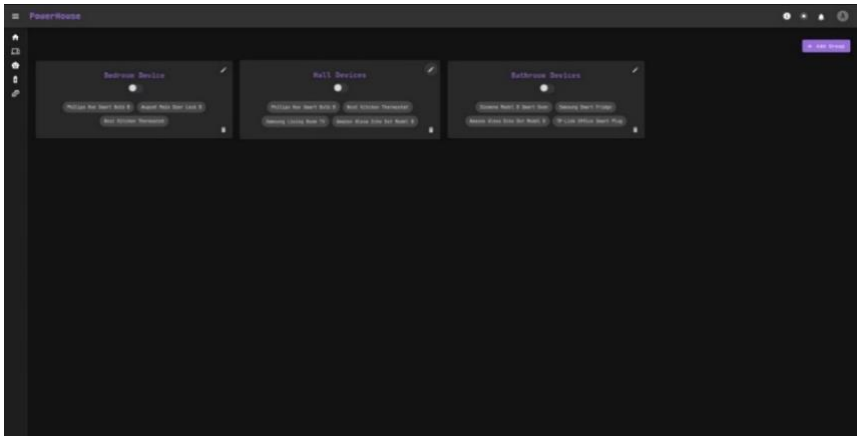
VI. Energy Analytics

The following tab depicts the analytics of the energy usage for the respective smart home. A responsive and dynamic pie chart is displayed, with hover-able properties, showing the amount of energy used per active device. Similarly, a line graph is shown to the right of the pie chart, showing the overall monthly energy usage of the smart home, accompanied by a user-set energy goal for analytical purposes. Further supplied is the Analytics Report generation, which generates a report between dates the user can set, showing detailed information about the energy usage per device, average energy usage, and energy tips based on how much energy has been used.



VII. Device Groups

The final section of the application is the Device Groups tab, where the user can compartmentalize some set of available devices to the user for ease of access. The user can add a group of their choice by clicking the [+ Add Group] button, which selects one or more available devices and groups them up as shown below. Through this, the user can choose to enable or disable the grouped-up devices with one push of a button.



4. Marketing Analysis

4.1 Summary

PowerHouse offers real-time energy monitoring, device automation, and sustainability insights to improve smart home energy management. With a growing emphasis on affordable and energy-efficient solutions, it positions itself as a user-friendly program that caters to a diverse range of households.

Through focused marketing initiatives, this marketing strategy seeks to:

- Increase user adoption and brand exposure.
- Make a difference through innovative energy-saving solutions.
- Highlight PowerHouse's distinctive qualities, including its user-friendly interface, sustainability features, and energy insights.
- Encourage consumer interaction through influencer collaborations, social media, and instructional materials.

This part of the report presents a thorough marketing plan to establish PowerHouse as a pioneer in the smart home sector. We will examine the existing marketing environment to optimize user engagement, raise brand awareness, promote long-term revenue growth, pinpoint important opportunities, and execute focused marketing campaigns.

4.2 Primary Objectives

1) **Energy Optimization & Efficiency**
   a) Monitor and analyse real-time energy usage
   b) Reduce unnecessary power consumption
   c) Provide AI-driven recommendations for efficiency
2) **Smart Automation & Control**
   a) Enable remote control via apps or voice assistants
   b) Automate schedules based on user behaviour
   c) Integrate seamlessly with other smart home devices

3) **Cost Savings & Affordability**
   a) Help users' lower electricity bills through optimized usage
   b) Offer transparent insights into energy consumption patterns
   c) Provide budget-friendly solutions with long-term ROI
4) **User-Friendly Experience**
   a) Intuitive interfaces for easy setup and control
   b) Simplified modes for non-tech-savvy users
   c) Customization features for personalized preferences

4.3 Product Background

- PowerHouse is an intelligent smart home energy management platform designed to enhance efficiency and sustainability. The app allows users to monitor real-time electricity consumption, track energy generation, and receive personalized recommendations to reduce costs.
- The app is designed to work seamlessly across devices, including computers, tablets, smartphones, and web browsers, making it accessible to all users.
- Its automated features and adjustable settings, which adjust energy consumption according to user preferences and daily routines, are a major selling point. Through clever power consumption optimization, it helps consumers save electricity costs and increase efficiency.
- Additionally, PowerHouse provides comprehensive energy statistics and insights, giving users in-depth data on their use patterns. The app also places a high priority on data security, protecting user information with sophisticated encryption and privacy protections.

4.4 Location

- The United Arab Emirates (UAE) presents a highly favourable environment for the launch of PowerHouse due to its commitment to sustainability.

- The UAE Net Zero by 2050 strategic effort lays out the country's ambitious plans to reach net-zero emissions by that year. This plan prioritizes the development of new technologies backed by research and development, and it is in line with international efforts to minimize temperature rise. To satisfy the nation's growing energy needs, the revised UAE Energy Strategy 2050 also calls for investing AED 150–200 billion by 2030 and tripling the share of renewable energy. These programs highlight the country's commitment to technical innovation and environmental growth, fostering an atmosphere that is favourable for energy management systems such as PowerHouse.
- The smart home market in the United Arab Emirates is expanding significantly due to consumer preferences and technological advancements. Given the high level of consumer interest in incorporating smart technologies into their homes, the market is expected to grow to significant sizes in the upcoming years. This growth is further supported by the UAE government's active promotion of smart city initiatives and investment in digital infrastructure, fostering an environment conducive to the adoption of smart home technologies (Advisors).

  People in the UAE who are tech-savvy are quite prepared to spend money on new technologies to improve their living areas. Smart home solutions are also being used because of government programs that support sustainable living and smart cities. PowerHouse is positioned favourably in the market thanks to its customer readiness and commitment to lowering carbon footprints.

4.5 Statistics and Market Size

The global smart home market is projected to reach USD 122.9 million by 2029, growing at a CAGR of 9.37% from 2025 to 2029. The increased adoption of IoT devices and smart home automation is driving this rapid expansion.

I. Energy Management

The smart energy management market is expected to grow from an estimated USD 49.01 billion in 2025 to a projected USD 84.34 billion by 2029, achieving a compound annual growth rate (CAGR) of 13.8%. This growth is driven by rising energy costs, the integration of renewable energy sources for power generation, and stringent government regulations aiming to reduce carbon footprints worldwide.

Target Demographics:

- Homeowners: Nearly 60% of homeowners are interested in smart home energy solutions (Sukys,2025).
- Tech Enthusiasts: Over 60% of users are willing to invest in automation tools.

- **Sustainability Advocates:** Consumers are willing to spend an average of 9.7% more on sustainably produced or sourced goods (PricewaterhouseCoopers).

II. Marketers

The digital advertising market was estimated at $734.24 billion in 2024 and is projected to touch $1426.63 billion in 2029 at a compound annual growth rate (CAGR) of 14.0%. This forecast can be attributed to the rise in advertising expenditure, increase in e-commerce penetration, and growth in social media usage (Company).

III. Marketing Budget Trends

Major trends in the forecast include an increased focus on strategic collaborations and partnerships to improve financial strength, usage of AI-powered apps for digital advertising, and the use of digital out-of-home (DooH) advertising campaigns to deliver advertising messages to target audiences.

4.6 Understanding of Beneficiaries and Stakeholders

I. Elderly Users

- **Simplified User Interface:** The elderly may not be as tech-savvy and might struggle with complex apps. To address this, Powerhouse will feature an intuitive design with large fonts and easy navigation.
    - Impact: This will make the app more user-friendly and accessible, encouraging adoption and reducing frustration.
- **Emphasize Safety and Convenience:** Many elderly individuals prioritize safety and convenience. Powerhouse should highlight how it helps users manage their energy usage effortlessly and safely.
    - Impact: Focusing on safety features, such as reminders to turn off appliances or automated energy-saving options, will enhance the app's appeal and provide peace of mind.
- **Offer Personalized Assistance:** Elderly users may need more personalized guidance. Offering support via phone or video calls can help them feel confident in using the app.
    - Impact: Providing accessible customer support ensures that users feel supported throughout the app's onboarding process and ongoing use.

- **Leverage Word-of-Mouth and Caregivers:** Many elderly individuals rely on caregivers or family members for advice on technology. Engaging caregivers or family members to promote the app will be crucial.
    - Impact: Encouraging family members to recommend the app will increase trust and adoption, as the elderly are more likely to trust suggestions from loved ones.
- **Highlight Financial Benefits:** Many elderly individuals are on fixed incomes. Powerhouse should emphasize how the app helps reduce electricity bills and saves money over time.
    - Impact: Promoting financial savings will resonate with elderly users, encouraging them to use the app to manage household expenses effectively.

II. Marketers

- **Promote Trust and Security:** The elderly may have concerns about privacy and security. Powerhouse should prioritize transparent communication about data security and privacy measures.
    - Impact: Ensuring elderly users feel confident that their personal information is protected will increase their trust in the app.
- **Use Traditional and Digital Media:** Not all elderly individuals are active on social media. A combination of online ads, email newsletters, and traditional marketing channels like TV or radio will be necessary to reach this audience.
    - Impact: Using both digital and traditional media will broaden the reach and ensure that the app engages with a diverse audience, including those who prefer more conventional forms of communication.

III. Stakeholders

The internal stakeholders include the PowerHouse employees, the development team behind the web application, the marketing team, and the client for whom PowerHouse was developed.

IV. Our Target Audience

Our target audience is the elderly, ensuring the website is accessible, easy to use, and designed with their needs in mind.

Core Value Proposition:

- Helps users monitor and optimize home energy usage

- Provides real-time analytics on energy consumption
- Offers automation features for improved efficiency
- Easy-to-use interface accessible from any device

4.7 Unique Selling Point

Powerhouse's unique selling points include:

- Real-Time Energy Tracking: Provides users with instant insights into their energy consumption, helping them monitor and optimize usage efficiently.
- Smart Automation Settings: Allows users to set custom automation rules for appliances and lighting, reducing unnecessary power consumption and maximizing convenience.
- Cost-Saving Insights: Analyses energy usage patterns and offers personalized recommendations to help users cut down on electricity bills.
- Interactive Dashboard: A sleek, high-contrast interface designed for readability, providing a user-friendly experience with easy navigation and data visualization.
- Smart Notifications & Reminders: This feature alerts users about peak energy usage, suggests off-peak operating hours, and sends reminders to turn off unused appliances.

4.8 Efficient Strategies for PowerHouse's Product Placement

I. Senior-Friendly Digital Presence & Marketing Webpage

- Develop an accessible, easy-to-navigate advertising web page designed specifically for older users.
- Use high-contrast visuals, large fonts, and simple layouts to enhance readability.
- Include step-by-step guides, demo videos, and a contact section for direct support.

II. Targeted Social Media Campaigns

- Focus on Facebook and Instagram, the most popular platforms among older adults.
- Share short videos, infographics, and user testimonials demonstrating PowerHouse's energy-saving features.
- Engage users with energy-saving tips and interactive content.

III. Community & Offline Engagement

- Distribute flyers and posters in libraries, senior centres, and local communities.
- Place hoardings on public transport and community spaces to increase visibility.

IV. YouTube & Video Marketing

- Create a professional demo video showcasing PowerHouse in action.
- Post tutorials on energy conservation tailored to older audiences.
- Feature real user stories and case studies to build credibility.

V. Survey-Driven Content & Support

- Conduct Google Forms surveys to understand senior users' biggest energy-saving challenges.
- Use insights to create relevant blogs, newsletters, and feature updates.
- Offer personalized support via phone or email to assist users in optimizing PowerHouse.

4.9 Competitors

- **Google Nest:** A smart home ecosystem offering intelligent thermostats, security cameras, and energy management solutions. It provides automation features but requires Google's ecosystem for full functionality.
- **Amazon Alexa Smart Home:** Integrates with various smart devices to enable voice-controlled automation, energy monitoring, and scheduling. However, it relies heavily on Amazon's ecosystem.
- **Sense:** A home energy monitor that provides detailed real-time electricity usage insights. Unlike PowerHouse, it requires additional hardware installation.
- **TP-Link Kasa Smart:** A smart device brand offering energy-efficient plugs, bulbs, and switches, allowing users to monitor and control power consumption. However, it lacks a centralized energy-saving intelligence system.
- **Emporia Energy:** Specializes in home energy monitoring, providing users with in-depth data analytics on power usage. However, it requires external hardware for full functionality.

Features	PowerHouse	Google Nest	Amazon Alexa	Sense	TP-Link Kasa Smart	Emporia Energy
Comprehensive Device Management	Yes	Yes	Yes	Yes	Yes	Yes
User-Friendly Access and Hierarchy	Yes	No	No	No	No	No
Advanced Energy Monitoring and Insights	Yes	Yes	Yes	Yes	Yes	Yes
Automation and Smart Scheduling	Yes	No	No	No	No	No
User Experience and Accessibility	Yes	No	No	No	No	No
Security & Compliance	Yes	No	No	No	No	No

4.10 SWOT Analysis

Strengths:

- Comprehensive Smart Home Integration: PowerHouse offers centralized control over a wide range of smart home devices, eliminating the need for multiple apps.
- Energy Consumption Insights: Provides real-time energy tracking, allowing users to monitor and optimize their usage.
- User-Friendly Interface: Simple and intuitive design accessible across different devices.
- Gamification Features: Encourages energy-saving habits through achievements and goal setting.

- Compatibility with Multiple Devices: Works with various Wi-Fi-enabled home devices, making it versatile.

Weaknesses:

- Dependence on Internet Connection: Requires a stable network for optimal performance.
- Limited Market Awareness: As a new entrant, PowerHouse needs strong branding and visibility.
- Potential Integration Issues: Compatibility challenges with different smart home brands and their protocols.

Opportunities:

- Growing Smart Home Market: Increasing adoption of IoT devices and home automation.
- Rising Energy Costs: More homeowners seek energy-efficient solutions to reduce bills.
- Sustainability Trend: Strong appeal to environmentally conscious consumers.
- Government Incentives: Potential for support through energy efficiency initiatives and rebates.

Threats:

- Competition from Established Brands: Companies like Google Nest and Amazon Alexa dominate the market.
- Cybersecurity Concerns: Smart home applications must ensure strong security measures to gain consumer trust.
- Regulatory Challenges: Data privacy laws and energy regulations may impact operations.

4.11 Marketing Goals

I. Awareness and Brand Positioning

- Establish PowerHouse as a leading solution for smart energy management.
- Use social media campaigns and content marketing to highlight key benefits.

II. Customer Acquisition

- Implement targeted digital advertising on Google and social platforms.
- Offer promotional incentives such as free trials and referral bonuses.

III. Customer Retention and Engagement

- Develop an email newsletter with energy-saving tips and feature updates.
- Implement a reward system for continued app usage and engagement.

IV. Partnerships and Collaborations

- Partner with smart device manufacturers for bundled offerings.

- Work with sustainability-focused organizations to reach eco-conscious consumers.

V. Primary Beneficiaries

- Homeowners: Seeking cost savings and improved energy efficiency.
- Tech Enthusiasts: Interested in smart home automation and IoT.
- Environmentally Conscious Consumers: Looking for sustainable solutions to reduce their carbon footprint.

VI. Stakeholders

- Investors and Venture Capitalists: Interested in the smart home and energy-saving market potential.
- Device Manufacturers: Potential partnerships for device integration.
- Government and Energy Regulators: May provide incentives for energy-efficient technologies.

4.12 PESTLE Analysis for PowerHouse

To assess the external factors influencing PowerHouse's growth and operational landscape, a PESTLE (Political, Economic, Social, Technological, Legal, and Environmental) analysis provides valuable insights.

Category	Description
Political 🟡	Stricter energy regulations and UAE Vision 2030 promote the adoption of smart energy solutions.
Economic 🟠	Rising energy costs drive demand for tools that help reduce energy consumption and lower utility bills.
Social 🟣	Growing environmental awareness increases consumer interest in energy conservation and eco-friendly solutions.
Technological ⚫	Advances in smart home technology enable better energy management and compatibility with more devices.
Legal 🔵	Compliance with data protection laws (e.g., UAE PDPL) ensures user privacy and security.
Environmental 🟢	The shift to renewable energy creates a need for tools to efficiently manage and optimize energy use.

4.13 Marketing Mix (7Ps)

PowerHouse employs the 7Ps framework to create a comprehensive strategy that enhances user engagement and meets customer needs.

I. Product

PowerHouse offers a user-friendly app that connects with various smart home devices, providing features such as real-time energy tracking, AI-powered recommendations, and customizable automation. Regular updates ensure security and compatibility.

II. Place

PowerHouse is accessible via multiple digital channels:

- **App Stores:** Available on the App Store and Google Play.
- **Web Access:** Compatible with web browsers for smartphones, tablets, and desktops.

III. Promotion

A multi-channel promotional strategy includes:

- **Social Media Campaigns:** Engaging content on Instagram, Facebook, and LinkedIn.
- **Influencer Collaborations:** Partnerships with tech influencers and eco-conscious advocates.
- **Content Marketing:** Blogs, tutorials, and newsletters to enhance user understanding.
- **Paid Advertising:** Targeted ads on digital platforms boost app downloads and usage.

IV. People

PowerHouse's team ensures seamless user experiences through:

- **Customer Support:** Live chat, email, and phone support.
- **Technical Team:** Regular updates and feature enhancements.
- **Marketing Specialists:** Brand campaigns and engagement strategies.

V. Process

PowerHouse simplifies energy management with:

- **Easy Onboarding:** Step-by-step setup guides.
- **Automation and Insights:** Personalized optimization recommendations.
- **Regular Updates:** Ongoing feature and security improvements.

VI. Physical Evidence

PowerHouse builds user trust through:

- **User Testimonials:** Real-life stories showcasing energy savings.
- **Tutorials and Guides:** Online resources and in-app tips.
- **Customer Reviews:** Positive feedback from users highlighting effectiveness.

By addressing these key elements, PowerHouse's marketing mix aims to drive growth, enhance user satisfaction, and strengthen its market position.

5. System Testing

This section details the testing and continuous integration/continuous deployment (CI/CD) processes used to maintain code quality and adherence to project specifications, enabling early detection and resolution of potential issues before production deployment.

5.1 Unit Testing

I. Backend Testing

Our team at PowerHouse Inc has been carrying out unit tests to ensure that the product is working properly and is staying consistent with our existing standards. Here are the tests we have implemented for our backend:

```
Run tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
1  ▶ Run pytest backend/tests
11 /opt/hostedtoolcache/Python/3.12.9/x64/lib/python3.12/site-packages/pytest_asyncio/plugin.py:207: PytestDeprecationWarning: The configuration option "asyncio_default_fixture_loop_scope" is unset.
12 The event loop scope for asynchronous fixtures will default to the fixture caching scope. Future versions of pytest-asyncio will default the loop scope for asynchronous fixtures to function scope. Set the default fixture loop scope explicitly in order to avoid unexpected behavior in the future. Valid fixture loop scopes are: "function", "class", "module", "package", "session"
13
14   warnings.warn(PytestDeprecationWarning(_DEFAULT_FIXTURE_LOOP_SCOPE_UNSET))
15 ============================= test session starts ==============================
16 platform linux -- Python 3.12.9, pytest-8.3.4, pluggy-1.5.0
17 rootdir: /home/runner/work/Production/Production
18 plugins: anyio-4.8.0, asyncio-0.25.3, mock-3.14.0
19 asyncio: mode=Mode.STRICT, asyncio_default_fixture_loop_scope=None
20 collected 27 items
21
22 backend/tests/test_automations.py .......                             [ 25%]
23 backend/tests/test_devices.py ....                                    [ 40%]
24 backend/tests/test_energy.py ..........                               [ 77%]
25 backend/tests/test_groups.py ......                                   [100%]
26
27 ============================== 27 passed in 0.14s ==============================
```

II. Frontend Testing

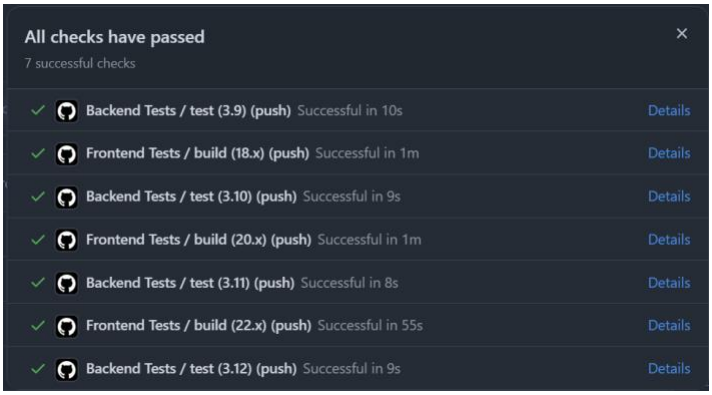
Our team for Frontend Testing has implemented compilation testing to ensure all our code for frontend compiles when we push it to production and to ensure there are no errors in the code:

```
Run npm run build --if-present
4
5  > frontend@0.1.0 build
6  > cd frontend && npx next build
7
8  ⚠ No build cache found. Please configure build caching for faster rebuilds. Read more: https://nextjs.org/docs/messages/no-cache
9  Attention: Next.js now collects completely anonymous telemetry regarding usage.
10 This information is used to shape Next.js' roadmap and prioritize features.
11 You can learn more, including how to opt-out if you'd not like to participate in this anonymous program, by visiting the following URL:
12 https://nextjs.org/telemetry
13
14   ▲ Next.js 15.1.7
15
16   Creating an optimized production build ...
17 ✓ Compiled successfully
18   Linting and checking validity of types ...
19 ⨯ ESLint: Cannot serialize key "parse" in parser: Function values are not supported.
20   Collecting page data ...
21   Generating static pages (0/11) ...
22   Generating static pages (2/11)
23   Generating static pages (5/11)
24   Generating static pages (8/11)
25 ✓ Generating static pages (11/11)
26   Finalizing page optimization ...
27   Collecting build traces ...
28
29 Route (app)                              Size     First Load JS
30 ┌ ○ /                                   3.98 kB        109 kB
31 ├ ○ /_not-found                          983 B          107 kB
32 ├ ○ /automations                         46.3 kB        244 kB
33 ├ ○ /dashboard                           10.7 kB        301 kB
34 ├ ○ /devices                             6.64 kB        185 kB
35 ├ ○ /energy                              16.4 kB        293 kB
36 ├ ○ /groups                              4.27 kB        189 kB
37 ├ ○ /login                               138 B          106 kB
38 └ ○ /users                               4.03 kB        181 kB
39 + First Load JS shared by all            106 kB
40   ├ chunks/239-363e0b1e9629564c.js       50.7 kB
41   ├ chunks/c/78296f7-752f66213dad4a00.js 52.0 kB
42   └ other shared chunks (total)          1.97 kB
43
44
45 ○  (Static)  prerendered as static content
```

5.2 GitHub pipeline

I. Automated Frontend and Backend Testing

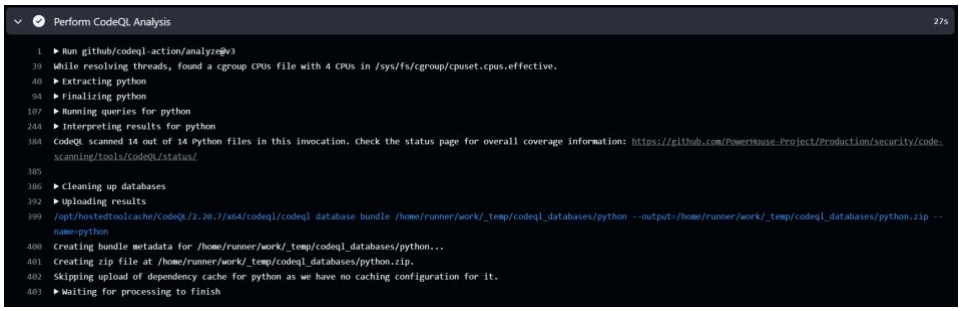
We have implemented pipelines for automating the frontend and backend testing whenever we push to our GitHub repository.



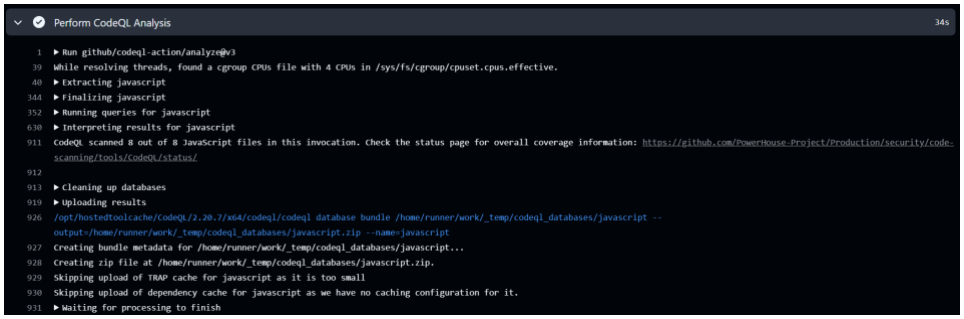
II. CodeQL Security Vulnerability Testing

CodeQL is a code analysis engine developed by GitHub, which we have used to automate security checks, identify vulnerabilities and errors in our code. CodeQL treats code as data, which then allows our development team to query the codebase to find potential issues with greater confidence than traditional static analyzers.

CodeQL Analyzer for Python:



CodeQL Analyzer for JavaScript



6. Project Evaluation

6.1 Organization

I. Group Organization

The PowerHouse team consists of eight dedicated members, each contributing uniquely to the successful completion of the project. At the outset, a SWOT analysis was conducted for every individual to allocate roles that align with their strengths. The team roles were flexible and adjusted between sprints as needed. The table below outlines each member's role and key contributions to the project.

Member	Formal Role	Contributions
Aditya Shibu	Project Manager	RW, BD, FD, P
David Farid	Technical Lead	RW, BD
Diing Juach	Scrum Master	RW, BD
Ann Erinjeri	Database Engineer	RW, DD, TT
Aoishi Chakravarti	Database Engineer	RW, M
Irsa Arshad	Database Engineer	RW, DD, TT
Saif Khalifa	Backend Developer	RW, P, M
Karim Jawhar	Frontend Developer	RW, P, M

Legend: RW – Report Writing, BD – Backend Development, FD – Frontend Development, DD – Database Development, TT – Testing Team, P – Prototyping, M – Marketing

II Group Collaboration

Throughout the project timeline, the team met regularly—at least once a week—to review sprint goals, discuss progress, and address any roadblocks. The Scrum Master coordinated meetings and ensured all voices were heard. The Project Manager engaged directly with the line manager to schedule consultations and collect feedback, which was critical to guiding development. Notes from each meeting were recorded and made accessible to those unable to attend. All tasks were tracked through Monday.com, and each was assigned realistic deadlines to maintain momentum toward completion.

III Online Tools

- WhatsApp: Used as the main channel for team-wide communication, including quick updates, deadline reminders, and discussions.
- Microsoft Teams: Enabled virtual meetings during holidays or in cases where in-person attendance was not possible.
- Notion: Served as a collaborative platform for documenting weekly meetings and team notes.
- GitHub: Managed our version control system, ensuring access to the latest codebase and preserving previous commits to maintain stability.

6.2 Implementation

I. Implementation Approach

The PowerHouse team followed the Scrum project management methodology. We worked in weekly or bi-weekly sprints to deliver iterative progress on the application. Each sprint produced functional increments that were evaluated, refined, and integrated into the broader system. This approach enabled continuous user-centered improvements, quick adaptation to changing requirements, and close collaboration between frontend, backend, database, and design teams.

II. Implementation Schedule

The implementation of the PowerHouse application was structured into development stages that focused on modular feature development. Our priority was to establish a reliable authentication system, followed by device control, energy monitoring, automation features, and UI enhancements:

- Weeks 1–2: Set up GitHub, establish front-end (React + Next.js) and back-end (FastAPI) structure
- Weeks 3–4: Build Firebase Authentication and user role system (super-user and sub-user)
- Weeks 5–6: Develop device pairing, command control APIs, and backend state tracking logic
- Weeks 7–8: Implement Firestore data logging, energy usage insights, and analytics dashboard (using Recharts & MUIX Charts)
- Weeks 9+: Add WebSocket for real-time monitoring, UI polish, bug fixing, usability testing

III Tools

a) React & Next.js:

React was selected for its component-based architecture, allowing developers to create isolated, reusable components. Next.js enhanced our performance by offering server-side rendering (SSR),

which provided faster page loads and better SEO. We also benefited from Next.js's file-based routing system and image optimization. These technologies ensured that the application remained responsive and functional.

b) Firebase (Authentication & Firestore):

Firebase was chosen for its seamless integration with React and real-time database capabilities. Firebase Authentication allowed us to implement secure sign-in using OAuth 2.0 with Google accounts, and it also handled user role assignments (super-user and sub-user). Firestore provided scalable NoSQL data storage and allowed our team to track real-time energy consumption and device status changes.

C) FastAPI:

FastAPI is a modern, high-performance web framework for building APIs with Python. It was used for handling backend logic, including device commands, data fetching, and analytics computation. Its built-in support for asynchronous operations made it suitable for managing multiple devices concurrently. FastAPI also helped us generate auto-documented API interfaces using FastAPI's auto documentation feature.

d) TailwindCSS:

TailwindCSS allowed us to design fast, flexible, and responsive user interfaces without writing extensive custom CSS. It enabled us to maintain consistent styling while customizing layouts for energy dashboards and device management screens. The use of utility-first classes also accelerated front-end development and facilitated collaboration between designers and developers.

f) Recharts & MUIX Charts:

To visualize energy consumption data in a user-friendly manner, we integrated Recharts and MUIX Charts libraries into our dashboard. Recharts offered component-based chart building using React, while MUIX Charts enhanced visual clarity with interactive components. Together, these tools allowed users to interpret daily, weekly, and monthly trends at a glance.

g) WebSockets:

WebSockets were used to provide real-time event communication between the client and server. This allowed users to receive instant feedback when devices were switched on/off or when usage

thresholds were crossed. This was crucial for implementing energy alerts and live monitoring.

h) GitHub:

GitHub served as our central repository and version control system. It allowed the team to collaborate efficiently using branches, pull requests, and issue tracking. The GitHub Actions feature was used to automate testing and deployments through our CI/CD pipeline, ensuring that code pushed to the main branch was stable and production ready.

6.3 Product

I. Requirements Assessment

The following table shows the implementation status of all functional and non-functional requirements defined in the Stage 1 PowerHouse requirements specification. As of the final project stage, all specified requirements have been implemented successfully.

Requirement ID	Requirement	Priority	Implemented
F-UR-1-1	Device Pairing (During Setup)	Must	Yes
F-UR-1-2	Device Pairing (After Setup)	Must	Yes
F-UR-1-3	Device Monitoring (After Setup)	Should	Yes
F-UR-2-1	Home Registration	Must	Yes
F-UR-2-2	User Login	Must	Yes
F-UR-2-3	User Hierarchy	Must	Yes
F-UR-2-4	User Profile	Should	Yes

F-UR-3-1	Remote Device Commands	Must	Yes
F-UR-3-2	Real-Time Feedback on Device Commands	Should	Yes
F-UR-3-3	Group Device Commands	Could	Yes
F-UR-4-1	Automation Scheduling	Should	Yes
F-UR-4-2	Condition-Triggered Actions	Could	Yes
F-UR-5-1	Energy Usage Reports (Super User)	Must	Yes
F-UR-5-2	Energy Usage Reports (Sub User)	Must	Yes
F-UR-5-3	Real-Time Energy Usage Reports	Should	Yes
F-UR-5-4	Energy Saving Recommendations	Could	Yes
F-UR-5-5	Energy Saving Achievements	Could	No
F-UR-5-6	Energy Saving Goals	Could	Yes
F-UR-5-7	Statistics Sharing on social media	Won't	Yes
F-UR-6-1	Energy Usage Reports (Super User)	Must	Yes
F-UR-6-2	Energy Usage Reports (Sub User)	Must	Yes
F-SR-1	Integration with Wi-Fi for device management	Must	Yes
F-SR-2-1	Device Pairing (During Setup)	Must	Yes
F-SR-2-2	Device Pairing (After Setup)	Must	Yes
F-SR-2-3	Device Status Monitoring	Should	Yes
F-SR-2-4	Device Status Updating	Should	Yes
F-SR-2-5	Device Status Alerts	Could	No
F-SR-3-1	Home Registration	Must	Yes
F-SR-3-2	User Login	Must	Yes
F-SR-3-3	User Hierarchy	Must	Yes
F-SR-3-4	Device Status Updating	Should	Yes

F-SR-4-1	Remote Device Commands	Must	Yes
F-SR-4-2	Real-Time Feedback on Device Commands	Should	Yes
F-SR-4-3	Group Device Commands	Could	Yes
F-SR-5-1	Automation Scheduling	Should	Yes
F-SR-5-2	Condition-Triggered Actions	Could	Yes
F-SR-6-1	Live Energy Consumption Data	Should	Yes
F-SR-6-2	Energy Usage Reports	Must	Yes
F-SR-6-3	Energy Usage Analysis	Must	Yes
F-SR-6-4	Energy Saving Recommendations	Could	Yes
F-SR-6-5	Energy Usage Achievements	Could	No
F-SR-6-6	Statistics Sharing on social media	Won't	Yes
F-SR-7	System-generated notifications for energy usage, device status, and commands	Should	Yes
NF-UR-1-1	Dashboard Customization	Could	No
NF-UR-1-2	Themes and Appearance	Could	Yes
NF-UR-1-3	Intuitive User Interface	Must	Yes
NF-UR-1-4	Tips and Tutorials	Should	Yes
NF-UR-1-5	Accessibility Settings	Should	No
NF-UR-2	Profile Lock with Passcodes	Could	Yes
NF-SR-1	Adjustable text size, color filters, high contrast mode, and screen reading	Should	No
NF-SR-2	Customization of UI elements	Could	No
NF-SR-3	Selection of preset themes and color palettes	Could	Yes

NF-SR-4	Easy-to-navigate UI, grouped functionalities, and accessible main functions	Must	Yes
NF-SR-5	Compliance with regional data regulations, encryption with libraries like BCrypt	Must	Yes

II. Product Uniqueness

PowerHouse stands out from existing smart home applications through its focus on sustainability, accessibility, and personalization. Unlike many smart home platforms that limit users to proprietary ecosystems or niche features, PowerHouse supports a wide range of Wi-Fi-enabled devices, offering centralized control through an intuitive web-based platform.

A key feature of uniqueness is the emphasis on energy awareness. Through visually rich dashboards built with Recharts and MUIX Charts, PowerHouse allows users to track their real-time energy consumption, set sustainability goals, and receive usage recommendations. Gamification elements like achievements for energy-saving milestones and the ability to share statistics on social media encourage user engagement and foster a sense of community around sustainable living.

PowerHouse also provides a flexible user hierarchy, allowing super users to manage multiple sub-users within a household. This makes the platform ideal for families, landlords, and shared living spaces. Its modular and scalable backend design ensures that new device types, automations, and insights can be added without overhauling the system.

III. Robustness

At the time of submission, the PowerHouse application functions without any critical errors or crashes. However, a few minor issues have been noted. These do not significantly impact on the functionality and can be resolved in future updates.

Section	Issue	Explanation

Energy Dashboard	Data Lag	Energy usage values take a few seconds to reflect due to backend processing time.
Sub-user Controls	Permission Glitches	Occasionally, sub-users are briefly shown devices they do not control when refreshing.
Device Pairing	Limited Auto-detection	Auto-discovery of devices works best with specific routers and may not support older IoT devices.

7. Usability Evaluation

Toward the end of the project, a usability evaluation was conducted using our application, hosted on a web browser. Participants were given realistic tasks, which highlight the core functionalities of the application. Feedback on user-experience was collected through both structured questionnaires and open discussion.

7.1 Test Plan

This document provides a detailed plan of Powerhouses' usability testing plan. The purpose of this test is to ensure that our application provides smooth-running user experience, which is intuitive, navigable, and interactable. This usability evaluation is crucial for making important observations about potential problems with the application, addressing feedback from user interaction, and refining the application to boost its usability.

The study will be operated on the actual application, demonstrating all its key functions and features to fully understand the application's user-friendliness, interactivity, and satisfaction.

This study is conducted under the guidance of our Line Manager Ms. Smitha Devidas Kumar as well as under the approval of our Project Coordinators Abrar Ullah and Ali Muzaffar. This study is entirely ethical and strictly for academic purposes.

7.2 Objectives

The purpose of this study is to determine which aspects of the application need to be further developed or updated, as well as to enhance the user's experience, guaranteeing that the product is in accordance with the user's requirements and anticipations.

In detail, our objectives encompass:

Collecting information from a diverse sample of participants to uncover input from a wider perspective.
Evaluating usability by assessing completion rates and average time taken (quantitative) and descriptive feedback (qualitative) for tasks that highlight the application's functionality.
Observing flaws of the application hinder usability with respect to specific tasks.
Addressing user critiques and using the recommendations collected through the evaluation to adjust and improve the application.

7.3 Methodology

I. Participants

The usability study is conducted as an expo-style session, with 15 testers, all aged 18 and above. When conducting usability testing, it is crucial to pick appropriate participants to obtain results that are dependable and actionable. Results from usability tests are more thorough and useful for system enhancements when they include a diverse variety of user experiences and preferences, thereby eliminating sampling bias. We took extreme importance in ensuring that the participants were from a diverse group by age, gender, occupation, field of occupation, and levels of computer literacy. The participants are responsible for providing transparent feedback regarding the usability and functionality of the application.

II. Training

Participants initially receive an explanation of the application's objective from the observers. If an aspect of the application makes them uneasy or uncertain, they are free to ask the facilitator questions. This will inform the team of any challenges the user encounters while utilizing the program or of any features that the user would be unaware of.

III. Procedure

The usability study for the PowerHouse application will be conducted in environments that are convenient for participants, ensuring their comfort and ability to focus throughout the testing process. Each session will be supervised by a team member who will guide participants through the study

while ensuring consistency in methodology across all sessions. The participants are first briefed about the study and the application. They then fill in the consent form and are assigned a unique identification participant number to ensure anonymization. They take the pre-testing questionnaire to garner sample demographics. Participants will then advance to application testing with the PowerHouse application, hosted on a PC browser, while tackling 15 predefined tasks specifically designed to evaluate the core functionalities and usability features of the application, during which the testers note down Task Completion Rate, Average Time Taken per task, and Problem Severity Rate. They then answer open-ended questions in the Testing Protocol highlighting the ease/difficulty of use, specific to each task. A post-testing questionnaire is also filled in by the participants to evaluate the app using a System Usability Scale (SUS), while also giving conclusive feedback. The tasks given to the testers represent common use cases derived directly from our requirements.

Task Scenarios Table:

Task ID	Task Scenario	Description	Requirements
T1	Login as a User	Log into the application using proxy Google account	F-UR-2-2
T2	Add new profile and set password	Make a new profile with a username and 4-digit password	F-UR-2-3
T3	Choose the Super-User profile	Select the super user profile from the existing profiles	F-UR-2-4
T4	Unlock Profile	Unlock the super user profile using a proxy PIN	NF-UR-2
T5	Turn off/on Philips Hue Smart Bulb B	Change the status of the device Philips Hue Smart Bulb B through remote device commands	F-UR-3-1
T6	Investigate daily and monthly energy usage.	View the graphs for energy usage making use of the real-time/daily/monthly toggles	F-UR-6 F-UR-5-3
T7	Change the name of August Main Door Lock A	Edit the name of the device August Main Door Lock A	F-UR-3
T8	Add / Delete Samsung Smart Fridge	Disconnect or re-connect the Device Samsung Smart Fridge	F-UR-1-2
T9	Turn on all the devices in bedroom	Give grouped device commands to the group Bedroom Device	F-UR-3-3
T10	Add a new automation	Make a new automation schedule and set its details	F-UR-4-1
T11	Generate a PDF of your choice for energy analytics	Make use of the Energy Analytics PDF Generator to make an energy analytics PDF of your choice in months, dates and devices	F-UR-5-1
T12	Switch to Dark Mode	Change the app's theme to dark mode	NF-UR-1-2

T13	Set a new energy goal	Create a new energy consumption goal for the household	F-UR-5-6
T14	Allocate devices to the new profile you created	Assign devices to your sub-user profile from the super-user profile	F-UR-2-3
T15	Check out the profile you created	Investigate the controlled access and features that your sub-user profile can access.	F-UR-2-4 F-UR-5-2 F-UR-6-2

IV. Consent Form:

Participants are given a consent form to be reviewed and signed in the beginning of the testing session. This document makes sure that the participants are fully informed about the objectives and nature of the study. It also clarifies that it is not their performance that is being evaluated, but the usability of the application. The participants are allowed to withdraw from the study at any time without providing a reason.

Link: Consent Form

V. Anonymization with Participant Number:

Every participant is assigned a unique identification number to ensure confidentiality of their personal data. Observers use this identification number throughout the session to record responses, observations, and feedback.

VI. Pretesting Questionnaire:

Participants will be instructed to complete a pre-testing questionnaire. The objective of this questionnaire is to collect demographic information and insights into the participants' prior experience with and knowledge of smart-home applications. Data such as age group, profession, familiarity with smart-home technology, and expectations for the PowerHouse application will be gathered. This information will provide a foundation for analyzing evaluation results and tailoring usability recommendations to different user groups.

Link: Pre-Testing Questionnaire

VII. Application Testing:

Participants will interact with the PowerHouse application to complete tasks such as logging in, creating a new profile, unlocking the super-user profile, activating devices, editing device names, investigating real-time, daily and monthly energy usage, unlocking profiles, generating energy

analytics PDF, switching to dark mode, adding automation schedules, enabling grouped device commands, setting energy goals, adding devices, and allocating devices to sub-user profile. Observers will track task completion rates, time taken, problem-severity rate and feedback to assess the app's usability and functionality, providing insights into its functional and non-functional requirements.

Link: Application

VIII. Test Protocol:

To maintain consistency across sessions, the facilitator will follow a standardized protocol. This includes providing a brief overview of the session structure and tasks without offering specific solutions. Participants will be encouraged to verbalize their thought processes while navigating the application. Observers will record interactions, suggestions, and challenges to inform future improvements. In consideration of privacy, no sessions will be digitally recorded; instead, all observations will be documented manually.

Link: Testing Protocol

IX. Post Testing Questionnaire:

At the conclusion of the session, participants will complete a post-testing questionnaire. This survey will capture feedback on user satisfaction, perceived ease of use, and the performance of specific features, including profile management, device management, device grouping, energy analytics, and automation schedules. The questionnaire will consist of ratings to quantify satisfaction levels and open-ended questions to gather detailed suggestions and highlight areas of frustration. The insights collected will be essential for refining the PowerHouse application in subsequent iterations.

Link: Post-Testing Questionnaire

7.4 Usability Goals

I. Completion Rate

The completion rate measures the percentage of users who successfully complete a task without critical errors. The goal is to ensure all users, regardless of background, can effectively interact with the system. For this test, the target completion rate is 100% for each critical task.

This is observed by the observer from the team during testing.

II. Average Time taken per Task

The average time taken by participants to finish each given task, on average, indicates the ease of use of the application. The more time a participant takes to finish a task, the more attention that needs to be drawn to enhance the said task-related area of the application. Our aim is for our participants to take an average of 5-15 seconds per task. This is observed by the observer from the team during testing.

III. Ease of Use Rate

On a scale of 1 to 5, with 5 representing the highest level of satisfaction, the users will rate their level of satisfaction and ease of usage with the application. Each page in our application aims to have a rating of 5.

This is computed during the Testing Protocol.

IV. Problem Severity Rate

The problem severity rate classifies and ranks usability problems according to how frequently they occur and how they affect testing. This strategy guarantees that urgent problems are attended to right away, while less relevant ones are dealt with gradually.

Two criteria are used to gauge each problem's severity:

1. **Frequency**: The proportion of participants that experience the problem.
    a. High: at least 30% of those who took part.
    b. Moderate: 11–29% of those who took part.
    c. Low: less than 10% of those involved.

2. **Impact**: How much the problem interferes with finishing a task.
    a. High: Blocks the completion of the task (critical error).
    b. Moderate: Takes more time or steps but doesn't stop you from finishing the task.
    c. Low: A minor annoyance that has no discernible impact on the task's outcome.

Every problem is categorized into one of four levels of severity:

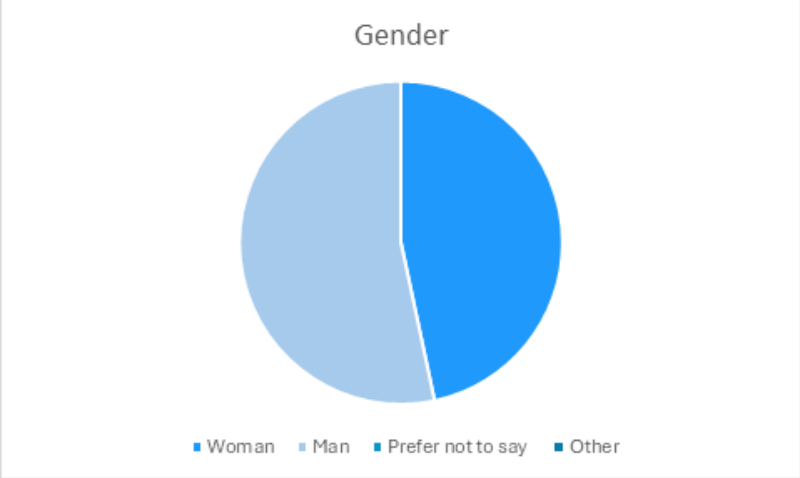
- Severity 1: High impact and high frequency (require immediate attention).
- Severity 2: High/moderate impact with moderate/low frequency.
- Severity 3: Moderate/minimal impact but high frequency.
- Severity 4: Minimal impact and low frequency.

7.5 Test and Questionnaire findings

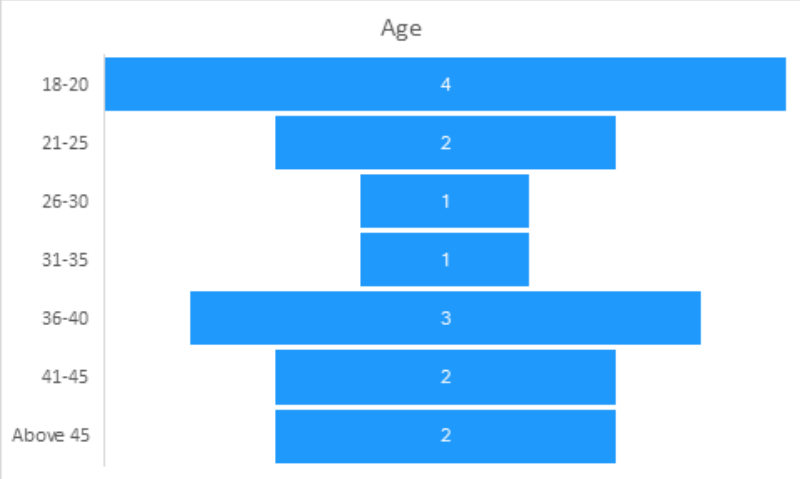
I. Demographics [Results from Pre-Testing Questionnaire]

The data presented below represents the 15 participants who took part in the Usability Study.

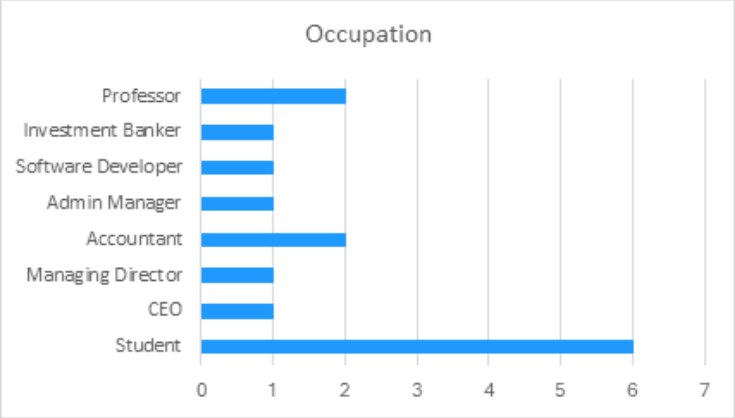
a) Gender:



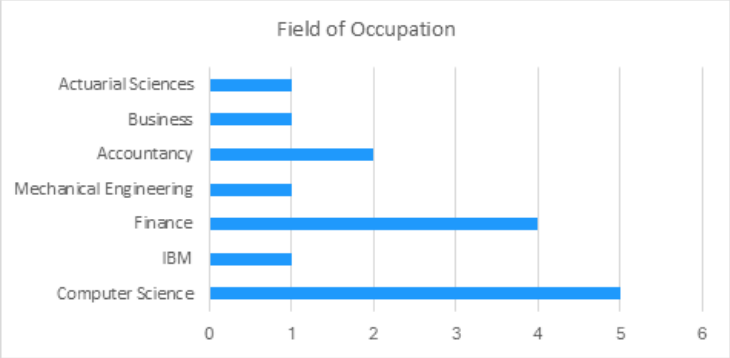
b) Age:



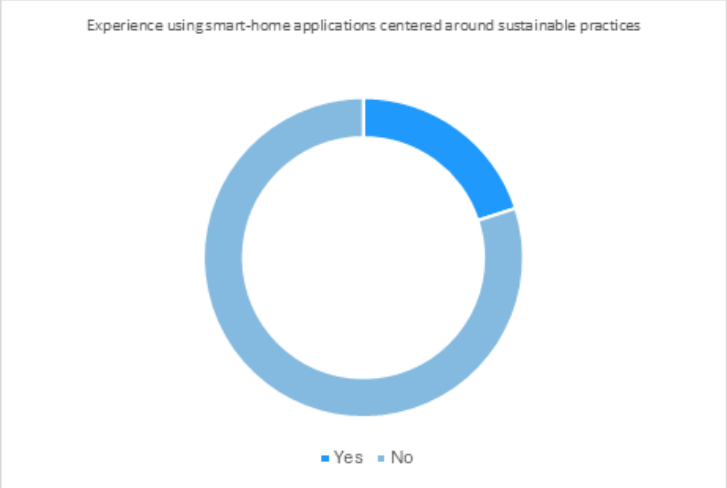
c) Occupation:



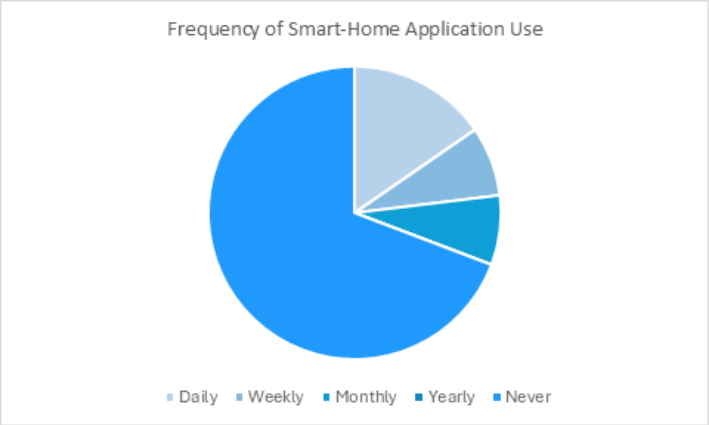
d) Field of Occupation:



e) Prior experience using a smart home application:

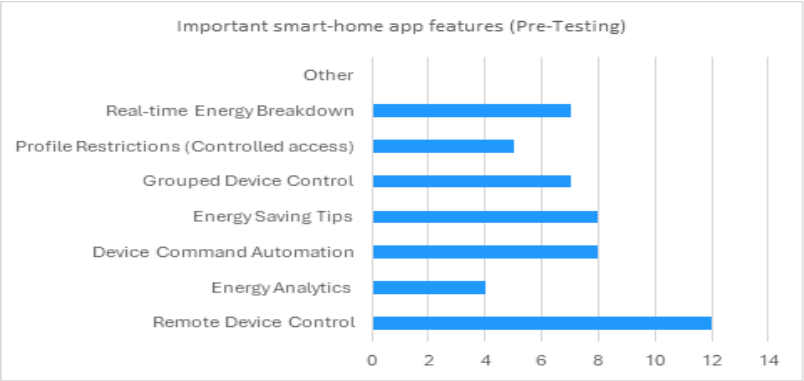


f) Frequency of using Smart home applications:



g) Rating on how satisfied the user is with the current smart-home applications available:

h) Smart-home application features deemed important by the participants:



II. Testing Findings and Comparison with Usability Goals [Results from Testing]

a) Completion Rates:

Task ID	Task Scenario	Completion Rate
T1	Login as a User [Login Page]	100%
T2	Add new profile and set password [Login Page]	100%
T3	Choose the Super-User profile [Login Page]	93.33%
T4	Unlock Profile [Login Page]	100%
T5	Turn off/on Philips Hue Smart Bulb B [Main Dashboard]	100%
T6	Investigate daily and monthly energy usage [Main Dashboard]	100%
T7	Change the name of August Main Door Lock A [Devices Page]	100%
T8	Add/Delete Samsung Smart Fridge [Devices Page]	100%
T9	Turn on all the devices in bedroom [Groups Page]	93.33%
T10	Add a new automation [Automations Page]	100%
T11	Generate a PDF of your choice for energy analytics [Energy Page]	100%
T12	Switch to Dark Mode [Main Dashboard]	93.33%
T13	Set a new energy goal [Energy Page]	100%
T14	Allocate devices to the new profile you created [Main Dashboard]	100%
T15	Check out the profile you created [Main Dashboard]	100%

This table shows the average completion rate per task.

The average overall completion rate is 98.67%.

Our usability goal was to attain an average completion rate of 100% for the tasks. We missed it by 1.33% and have honed the application to ensure 100% completion rates (See in Improvements).

b) Average Time taken per Task:

Task ID	Task Scenario	Time taken/s
T1	Login as a User [Login Page]	2.3
T2	Add new profile and set password [Login Page]	11.2
T3	Choose the Super-User profile [Login Page]	9.1
T4	Unlock Profile [Login Page]	6.6
T5	Turn off/on Philips Hue Smart Bulb B [Main Dashboard]	1.2
T6	Investigate daily and monthly energy usage [Main Dashboard]	20.1
T7	Change the name of August Main Door Lock A [Devices Page]	32.7
T8	Add/Delete Samsung Smart Fridge [Devices Page]	4.2
T9	Turn on all the devices in bedroom [Groups Page]	23.3
T10	Add a new automation [Automations Page]	12.9
T11	Generate a PDF of your choice for energy analytics [Energy Page]	17.7
T12	Switch to Dark Mode [Main Dashboard]	8.3
T13	Set a new energy goal [Energy Page]	19.8
T14	Allocate devices to the new profile you created [Main Dashboard]	21.7
T15	Check out the profile you created [Main Dashboard]	7.9

This table shows the average time taken per task.

The average time taken for all the tasks is 13.2666667 seconds.

Our usability goal was to attain an average of 5-15 seconds taken for the tasks. Our findings fall within the same range with some outliers (T6, T7, T9, T11, T13, T14). We have honed the application to reduce the average time taken for these tasks (See in Improvements).

c) Ease of Use rate:

Tasks in each Page	Rating (Highest to Lowest)
Log In Page	4.93
Automations Page	4.87
Devices Page	4.67

Groups Page	4.53
Energy Page	4.53
Main Dashboard	3.93

This table shows the ease-of-use rating per task.

The average ease-of-use rating over all the pages is 4.58.

This rating is lower than our desired ease-of-use rating of 5 and is 91.53% of the desired rating.

We have honed the application to increase the overall ease-of-use rating (See in Improvements).

d) Problem Severity Rating:

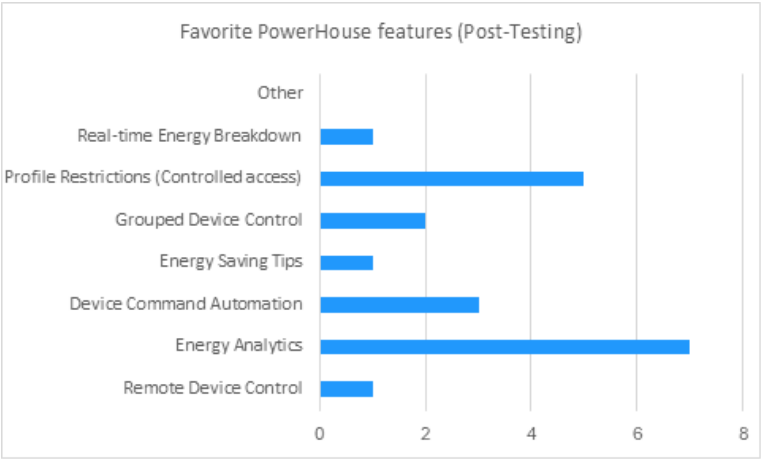
Problems encountered by the Participants during the Testing are documented through the Testing Protocol.

Problem ID	Problem Description (High to Low)	Impact of the Problem	Percentage who encountered the problem	Problem Severity Rate
P1	Information on the application's features is not integrated into the app.	High	40%	Severity 1
P2	Dashboard icons are not hoverable.	High	33.33%	Severity 1
P3	Buttons for the energy graph toggles are not big enough.	High	20%	Severity 2
P4	Device groups icon is ambiguous.	High	20%	Severity 2
P5	The house icon unintentionally logs the user out.	High	13.33%	Severity 2
P6	The password is not typable.	Moderate	13.33%	Severity 2
P7	The button to allocate devices to sub-users was not explicit enough.	Moderate	13.33%	Severity 2
P8	Profile tiles don't have the user-role mentioned explicitly.	Moderate	6.67%	Severity 4
P9	The energy analytics PDF generator doesn't accept all date ranges.	Moderate	6.67%	Severity 4
P10	There are limited color themes.	Low	6.67%	Severity 4

Studying the feedback collected from the participants in the study, we note the important problems with our application.

The table shows how we took note of the impact and the frequency of the problem to rank the problems in order of Problem Severity Rate.

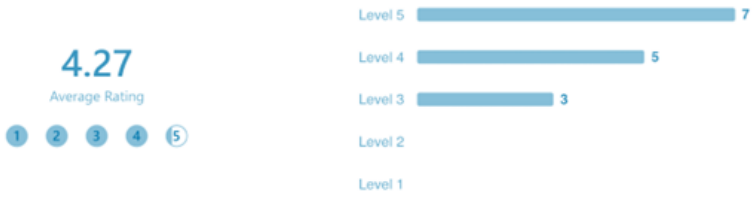
III. Favourite Application features and comparison with UR



This row chart shows the PowerHouse application features which the participant loved the most.

We compare the needs and wants of the users to the existing features of the application and find that it fulfils and exceeds the expectations of the test participants.

IV. Post-Testing System Usability Scale SUS Findings



The average rating on the application's intuitiveness is 4.27

The average rating on the application's design is 4.80

The average rating on the application's recommendableness is 4.67

As close as we are to our usability goals, this indicates the need for significant improvement for the application.

7.6 Improvements

Assessing our findings, we found some flaws in our application, which hinder user experience. These discoveries were reported to the backend and frontend team, and the problems were solved efficiently in the order of Problem Severity.

The improvements have been listed on the table below in detail with a solution to address these issues. Although some have been addressed, not all of them have been resolved in the current version of the application. These fixes and solutions may be implemented in the future if deemed necessary.

Problem ID	Problem Description (High to Low)	Problem Severity Rate	Fix
P1	Information on the application's features is not integrated into the app.	Severity 1	A downloadable user guide was integrated into the app which gave an in-depth description of each of the app's features.
P2	Dashboard icons are not hoverable.	Severity 1	The dashboard sidebar was made hoverable to make the application more navigable.
P3	Buttons for the energy graph toggles are not big enough.	Severity 2	All the buttons were made bigger in consideration of our major stakeholders (the elderly).
P4	Device groups icon is ambiguous.	Severity 2	A more intuitive icon was chosen for Device Groups.

P5	The house icon unintentionally logs the user out.	Severity 2	The redundant house icon was removed and instead a more useful switch profiles button was introduced.
P6	The password is not typable.	Severity 2	The password was made typable while also remaining compatible with key pressing.
P7	The button to allocate devices to sub-users was not explicit enough.	Severity 2	The button was relocated to the profile bar to make the action more executable.
P8	Profile tiles don't have the user-role mentioned explicitly.	Severity 4	All the profile tiles were assigned explicit user-roles in the User List.
P9	The energy analytics PDF generator doesn't accept all date ranges.	Severity 4	The PDF generator was made more flexible with date ranges.
P10	There are limited color themes.	Severity 4	This problem has yet to be fixed but is not deemed difficult to execute in coming iterations of the app.

This table shows the fixes and the order in which each problem was fixed.

7.7 Usability Conclusion

The PowerHouse usability study, which included 15 diverse participants, aged 18 and above, revealed important insights on user engagement, satisfaction, and functionality. The demographics revealed differences in age, occupation, tech knowledge, and gender. In terms of features, users favored energy analytics, automation, and grouped device control.

Participants completed 15 realistic activities based on core requirements, resulting in an average task completion rate of 98.67%, just 1.33% less than the 100% aim. The average time per task was 13.27 seconds, which met the usability guideline of 5-15 seconds; however, six tasks (T6, T7, T9, T11, T13, and T14) exceeded expectations slightly. The average ease-of-use rating was 4.58/5, which is just shy than the optimum score of 5, with the Main Dashboard scoring the lowest (3.93) and the Log In Page scoring the best (4.93).

Crucially, the survey found that users appreciated several of the fundamental features, particularly those related to energy monitoring, automation, and profile management. However, some UI/UX components, such as non-hoverable icons, insufficient toggle buttons, and a lack of in-app advice, posed hurdles to effective engagement. By addressing issues based on their severity, the development team was able to make significant adjustments that improved the overall user experience.

We have adopted an iterative approach to usability testing which demonstrates PowerHouse's commitment to developing user-centered solutions. The results of this review not only improved the present version of the program, but also offered a road map for future updates.

PowerHouse is now more approachable, user-friendly, and customized to meet the needs of users from a variety of backgrounds by incorporating user feedback and improving pain areas.

8. User Guide

8.1. Basic Terminologies

I. Dashboard

The main dashboard page where users can see all their energy usage, devices connected, automation schedules and device controls in a single page

II. Devices

The devices page shows you the controls for all devices, as well as options to edit the names and delete these devices from PowerHouse

III. Automation

The automations page lists all the automation schedules created, Automations are to mainly remotely control devices based on a trigger, such as at a certain time

IV. Energy

The energy page showcases the device wise breakdown of energy usage, and the monthly energy usage vs the energy goal set by the user. This is also the page where you will find the option to generate the Analytics PDF for sharing to social media

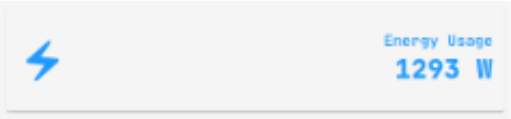
V. Groups

Groups help you create a group of devices that can be controlled together, such as turning a group of devices in your bedroom

VI. Users

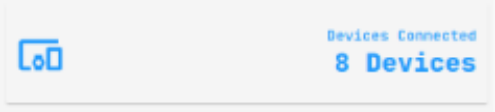
PowerHouse has 2 types of users, the Super User and the Sub User. The Super User is an admin who has access to all devices, and can control and allocate devices to sub users

8.2. Dashboard



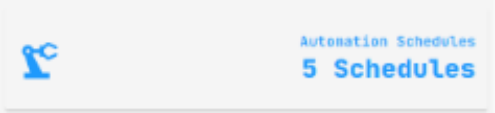
Real-time Energy Usage

Displays the real time energy usage being consumed by



All Connected Devices

Displays the number of connected devices in PowerHouse, clicking on this will take you to the Devices page



Automation Schedules

Displays the count of all automation schedules created in PowerHouse

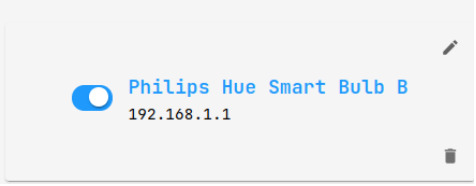
Device Control

Displays all the devices connected to PowerHouse along with the option to toggle their power status

Real-time, Daily and Monthly Energy Usages

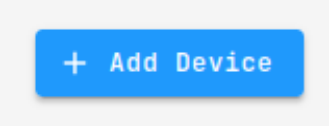
Toggle between real-time, daily and monthly energy Usage graphs

8.3. Devices



Device Card

Shows all the devices in the form of a card with the option to Edit their names and delete the device from PowerHouse



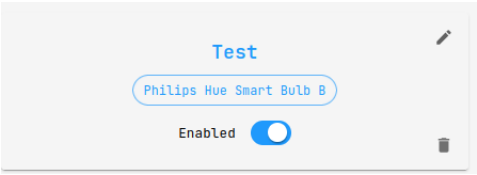
Add a new Device

The add device button shows all the devices that are not connected to PowerHouse

Add Device Dialogue

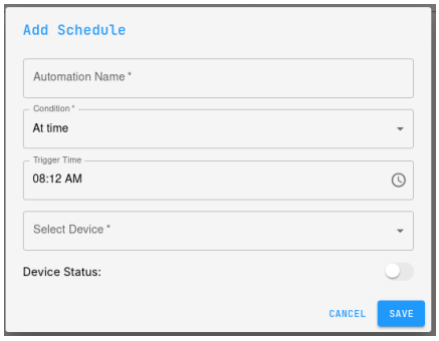
The add device dialogue shows a dropdown which allows you to select devices not connected to PowerHouse

8.4. Automations



Automation Card

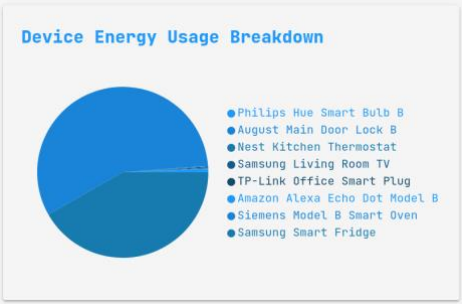
Shows all the automations as cards, with the option to edit, delete and deactivate



Add Automation Dialogue

The dialogue allows you to give the automation a name, set a condition and the device that the automation should affect, along with if the automation should turn the device on or off using the device status field

8.5. Energy

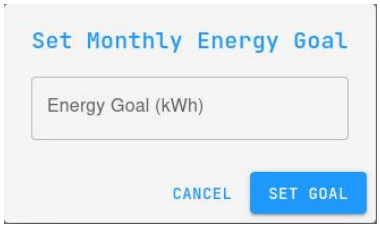


Energy Usage Breakdown

Provides detailed insights into the energy consumption of each connected devices.

Generate Energy PDF

The dialogue allows you to create a PDF summary of energy usage over a selected date range and devices.



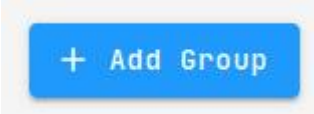
Set Monthly Energy Goal

The dialogue allows you to set a monthly energy goal.

Energy Goals

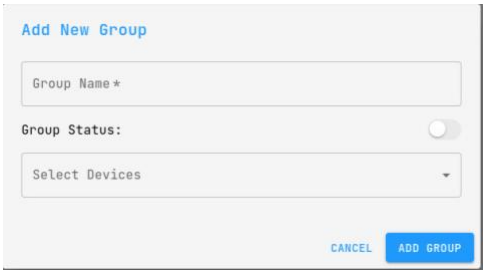
Allows you to track your progress toward your set

8.6. Groups



Add a new Group

The add group button lets you create a new group with selected devices of your choice.



Add Group Dialogue

The add new group dialogue allows you to name your group and select devices from those allocated to your profile using a dropdown menu.

Group Card

Shows all the groups as cards, with the option to edit group name, delete group and manage devices within the group.

8.6. Users

Add User Dialogue

Add new user dialogues allows you to add a new user and allocate devices to them using the dropdown menu, that shows all the devices that are connected to powerhouse.

Allocate Devices Dialogue

Allocate Devices Dialogue lets you assign devices to a user connected to the powerhouse.

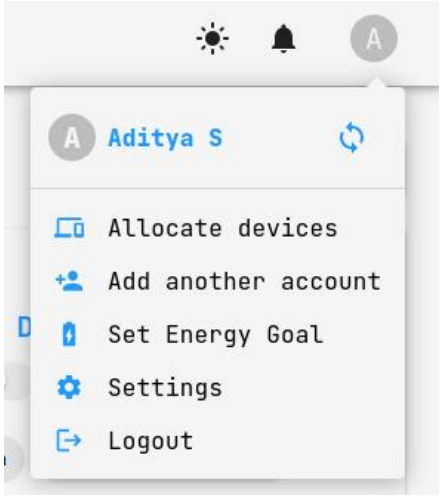
User Settings

User settings display your information, including the account you're logged in with, your account type, and all the devices allocated to your account.

8.6. Navigation

User Menu

User menu lets you access your profile and notifications, and switch between theme settings.

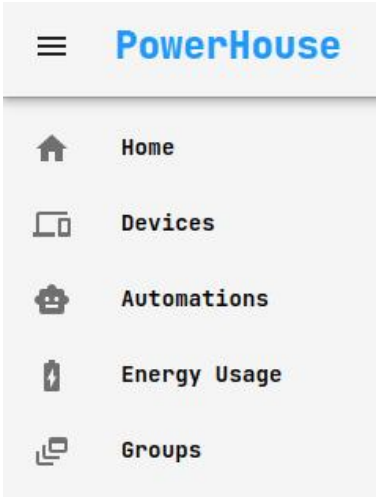


Profile Icon

Profile icon displays all the actions you can perform including allocating devices, adding another account, setting an energy goal, accessing settings, and logging out.

Sidebar Menu

Sidebar menu lets you easily navigate through powerhouse. It allows you to manage devices, groups and automations, as well as view energy usage.


PowerHouse
Home
Devices
Automations
Energy Usage
Groups

9.Appendix

9.1 Meeting Diary

Meeting #1 - 08/01/2025

Present: All Members

Absent: None

Meeting Duration: 1 hour 15 minutes

Meeting Notes:

- Discussed project scope and purpose.
- Assigned initial roles and responsibilities to team members.
- Reviewed functional and non-functional requirements.
- Outlined an initial risk analysis framework.
- Agreed on communication and project management tools.
- Set up GitHub repository for version control.

Next Steps:

- Begin drafting wireframes for the user interface.
- Identify the app's unique selling points (USP).
- Define key system architecture components.

Meeting #2 - 15/01/2025

Present: All Members

Absent: None

Meeting Duration: 1.5 hours

Meeting Notes:

- Reviewed and refined user stories based on different user types (Super-user & Sub-user).
- Discussed connection protocols for Wi-Fi-enabled devices.
- Evaluated various smart home device integrations and compatibility concerns.
- Confirmed Next.js as the front-end framework for development.
- Outlined database structure and data flow for device status monitoring.
- Assigned research tasks for API and backend service integration.

Next Steps:

- Assign tasks for database schema design.

- Research Firebase authentication methods.

Meeting #3 - 22/01/2025

Present: Aditya, Aoishi, Ann, Karim, Irsa, Saif

Absent: David, Diing

Meeting Duration: 1 hour

Meeting Notes:

- Reviewed wireframe drafts and gathered feedback for improvements.
- Discussed front-end development approach.
- Outlined API structure and endpoints for device control features.
- Allowed users to set custom schedules (e.g., "Turn off lights at 10 PM").
- Defined data storage requirements for energy usage tracking.
- Established security considerations for user authentication and data encryption.

Next Steps:

- Begin front-end development.

Meeting #4 - 30/01/2025

Present: All Members

Absent:

Meeting Duration: 2 hours

Meeting Notes:

- Reviewed front-end progress and identified required UI adjustments.
- Created a device management dashboard for users.
- Discussed integration of Firebase with the back-end system.
- Defined key API routes for device management and energy monitoring.
- Evaluated security protocols for user authentication.

Next Steps:

- Implement API endpoints for device pairing and status updates.

Meeting #5 - 06/02/2025

Present: Aditya, Ann, David, Diing, Karim, Saif

Absent: Irsa, Aoishi

Meeting Duration: 1 hour 45 minutes

Meeting Notes:

- Conducted a code review of front-end and back-end components.
- Reviewed the database schema for device data and user profiles.
- Tested initial API calls between front-end and back-end.
- Identified key performance optimization areas.
- Developed initial class diagrams to represent system architecture.
- Implemented a recurring schedule feature (daily, weekly, monthly).
- Planned the usability testing strategy.
- Developed real-time energy consumption tracking for smart devices.

Next Steps:

- Document database structure and API endpoints.

Meeting #6 [Line Manager] - 20/02/2025

Present: All members

Absent: None

Meeting Duration: 1 hour 10 minutes

Meeting Notes:

- Reviewed the use case diagrams to ensure all functional interactions are well-defined.
- Discussed how to align class diagrams with the database schema.
- Identified missing entities and relationships in the diagrams.
- Planned updates to the diagrams based on the Line manager's feedback.

Next Steps:

- Finalize and document the use case and class diagrams.
- Ensure consistency between system architecture and diagrams.

Meeting #7 - 27/02/2025

Present: David, Saif, Irsa, Aditya, Aoishi, Diing

Absent: Ann, Karim,

Meeting Duration: 55 minutes

Meeting Notes:

- Developed marketing strategy, including user personas and target audience.

- Created an initial branding concept (logo, color scheme, and typography).
- Discussed potential marketing materials (flyers, social media content).
- Planned upcoming feature demonstration for project advisors.
- Reviewed implementation progress and identified remaining features.
- Implemented downloadable energy reports in PDF format.
- Ensured reports are professionally formatted with clear visual elements.

Next Steps:

- Finalize branding and create marketing materials.
- Work on remaining feature implementations.

Meeting #8 - 05/03/2025

Present: All members

Absent: None

Meeting Duration: 1 hour 5 minutes

Meeting Notes:

- Finalized the user guide for the application.
- Reviewed and updated the test cases for the usability testing.
- Tested authentication flows and resolved login-related issues.
- Reviewed documentation progress.
- Assigned tasks for final report preparation.

Meeting #9 - 12/03/2025

Present: All Members

Absent:

Meeting Duration: 1 hour 20 minutes

Meeting Notes:

- Addressed final usability concerns.
- Included detailed breakdowns of energy consumption per device and time.
- Prepared some content for the final product demonstration.
- Conducted stress testing of the system.
- Assigned final debugging and optimization tasks.

Next Steps:

- Work on the final documentation.

Meeting #10 - 16/03/2025

Present: Aoishi, Irsa, Diing, Saif, Karim, Ann

Absent: Aditya, David

Meeting Duration: 2 hours

Meeting Notes:

- Finalized PowerHouse branding and marketing strategy.
- Completed all system testing and debugging.
- Updated project risk assessment based on system performance.
- Reviewed overall project progress against the initial timeline.

Meeting #11 [Line Manager] - 18/03/2025

Present: All members

Absent: None

Meeting Duration: 45 minutes

Meeting Notes:

- Presented the marketing website to the Line manager.
- Discussed concerns regarding the final Class diagram.
- Reviewed the social media accounts for marketing.

Next Steps:

- Finalize all marketing materials.

Meeting #12 - 20/03/2025

Present: Aoishi, Irsa, Saif, Karim, Ann, Aditya, David

Absent: Diing

Meeting Duration: 1 hour 10 minutes

Meeting Notes:

- Conducted a final technical review of the system.
- Discussed potential improvements for future iterations.
- Conducted a bug-check and system stability test.
- Tested PDF generation on various devices and screen sizes.
- Prepared for the final submission and presentation.

Next Steps:

- Complete any last-minute refinements to documentation.

Meeting #13 - 22/03/2025

Present: David, Ann, Irsa, Karim, Diing, Aditiya, Aoishi

Absent: Saif

Meeting Duration: 50 minutes

Meeting Notes:

- Finalized the system for demonstration.
- Ensured all documentation is ready for submission.

Next Steps:

- Wrap up project documentation and archive files.

Meeting #14 - 25/03/2025

Present: All members

Absent: None

Meeting Duration: 1 Hour

Meeting Notes:

- Conducted a trial run of the final demo presentation.
- Ensured all documentation aligns with submission guidelines.

Next Steps:

- Wrap up project documentation and archive files.
- Submit the final project and prepare for team evaluations.

9.2 Marketing Materials

Instagram link:

https://www.instagram.com/powerhousehwud/?utm_source=qr&igsh=dzg1aTlpanNhNDF4#

LinkedIn Link:

https://www.linkedin.com/company/officialpowerhouse/

YouTube Link:

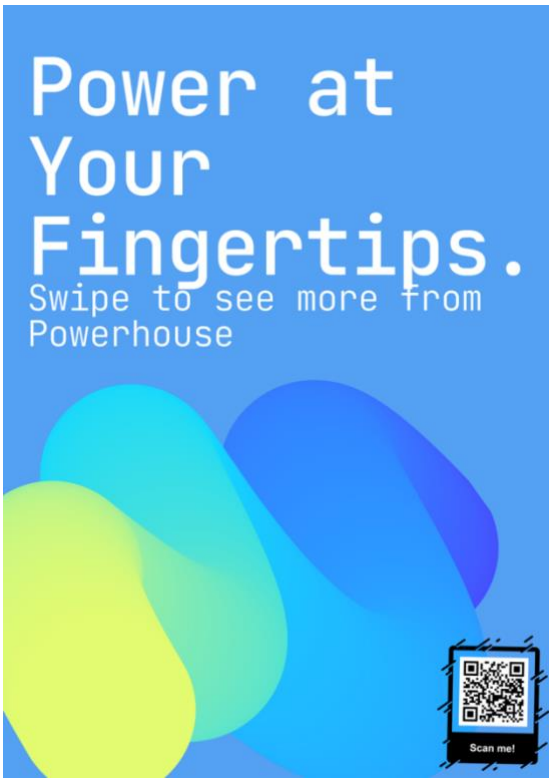
https://youtube.com/@powerhouse-hwud?si=Sv2VjKZbUhVnreVR

Facebook Link:

https://www.facebook.com/people/PowerHouse-Hwud/pfbid035ThqT7RY5LP3ebLXC7R815uh2aDPqZoH48dApxvaPnSu2hkKP9bW8sufsPpFAbW9l/

Social Media Post Materials

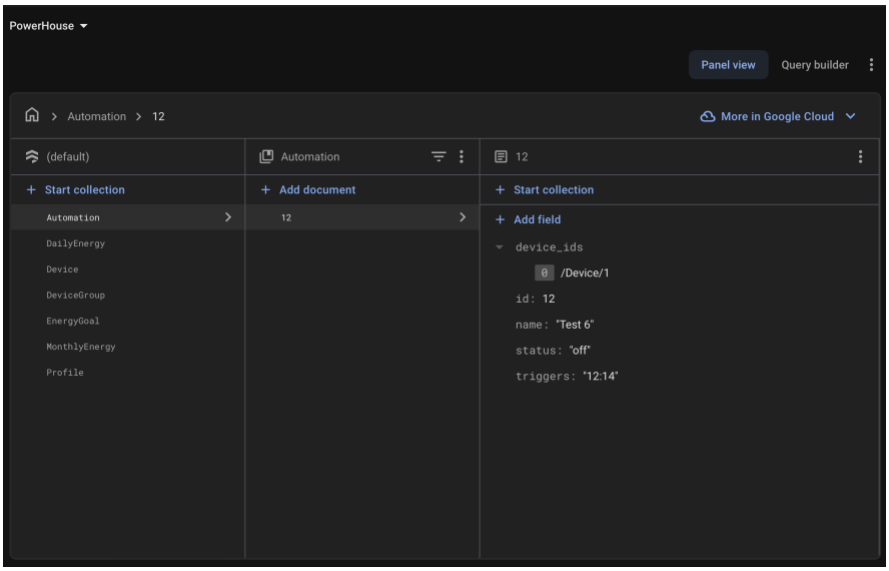
Posters


Power at Your Fingertips.
Swipe to see more from Powerhouse
Scan me!

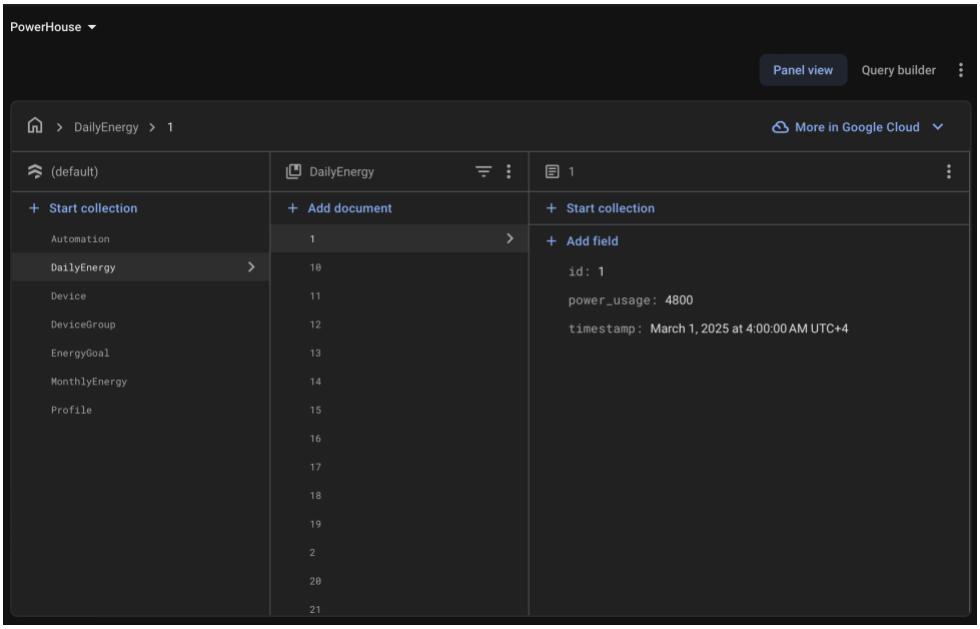
Ready to Save Energy!
Swipe up to Join us at Powerhouse
Scan me!

9.3 Firestore Database

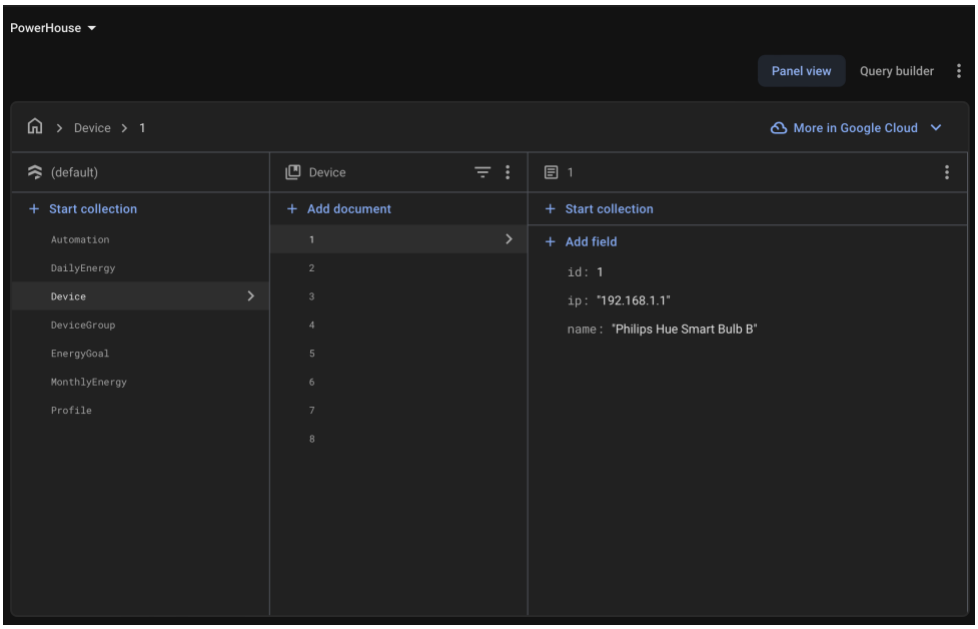
I. Automation Collection



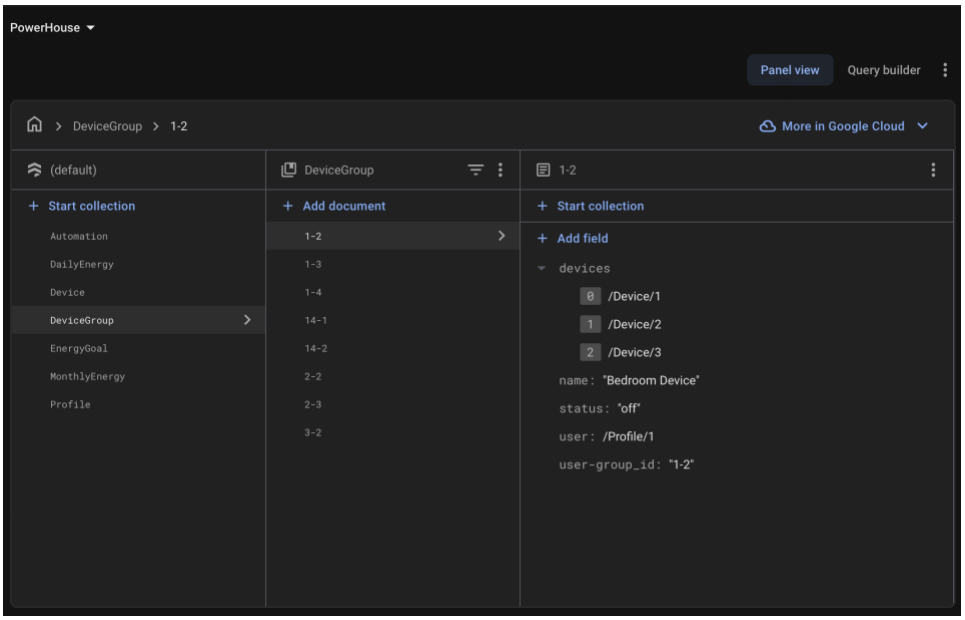
II. DailyEnergy Collection



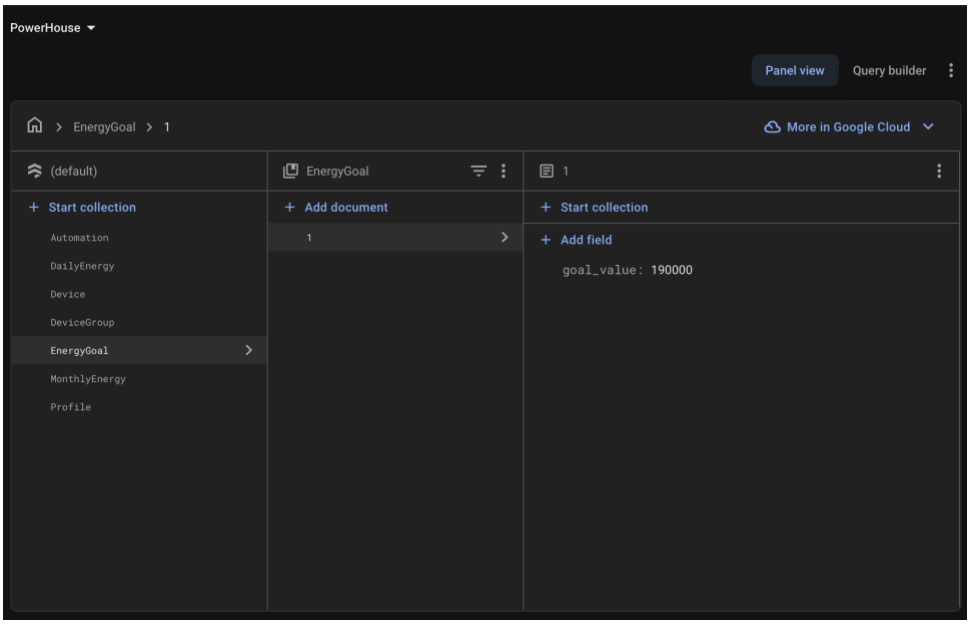
III. Device Collection



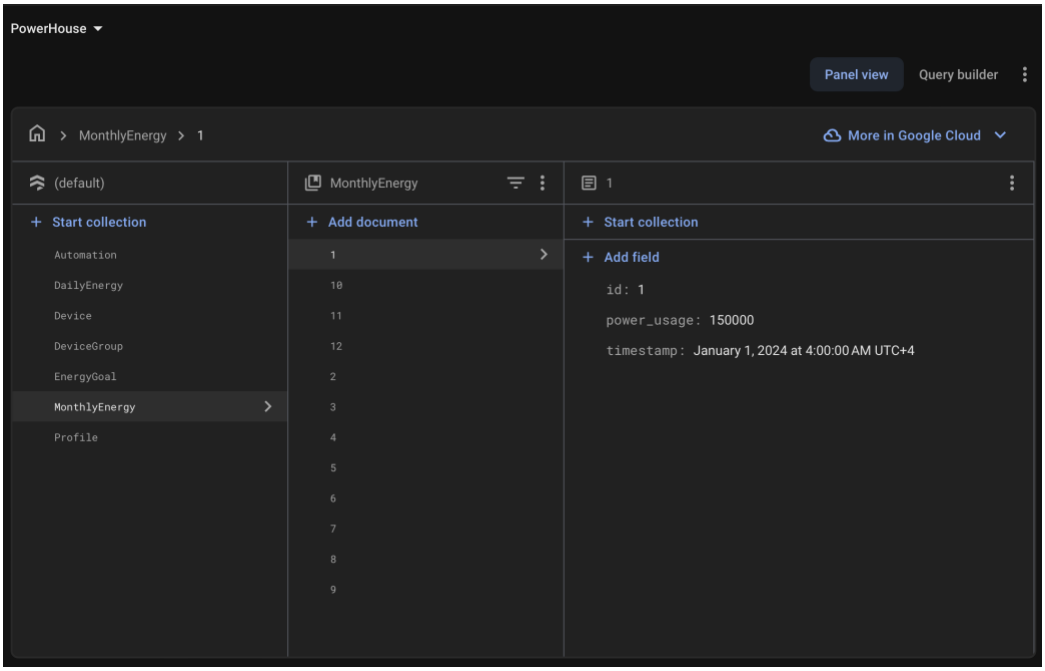
IV. DeviceGroup Collection



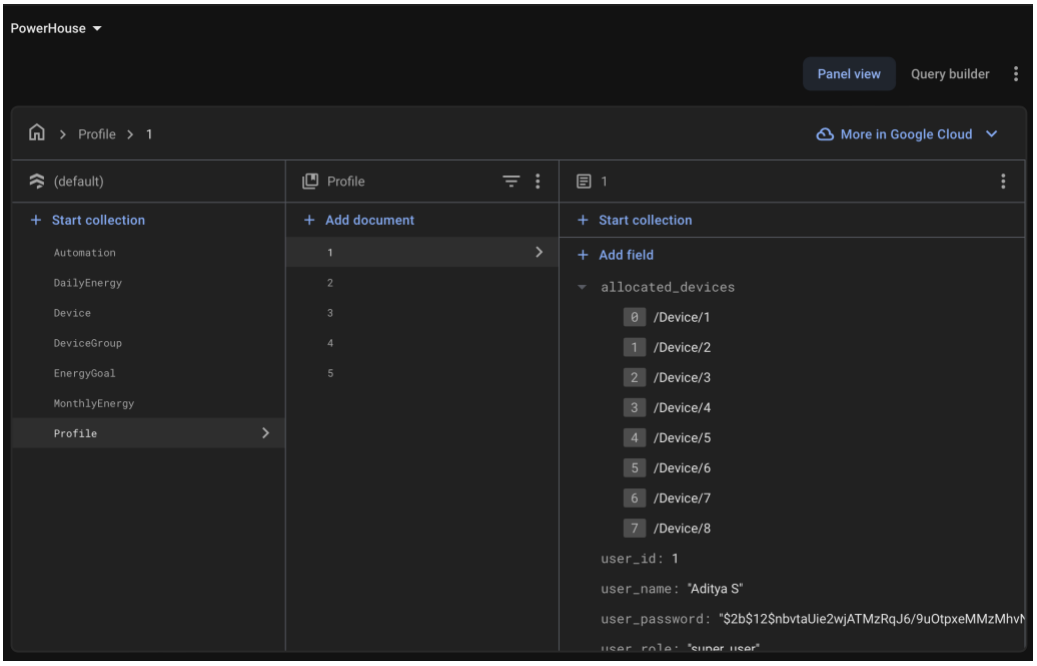
V. EnergyGoal Collection



VI. MonthlyEnergy Collection



VII. Profile Collection



9.4 Usability Study Consent Form

[STAGE 2] Consent Form for Usability Test Participants

PowerHouse
Heriot-Watt University

Principal Investigators
Aditya Shibu, David Farid, Aoishi Chakravarti, Ann Erinjeri, Irsa Arshad, Karim Jawhar, Diing Juach, Saif Khalifa

PowerHouse PowerHouse is our smart home application which aims to empower users to seamlessly control and automate their home environment, enhancing comfort, security and energy efficiency.

The objective of this study is to test the usability of our application PowerHouse. The study aims to evaluate the users' ease of browsing through and completing the task scenarios on the application. The qualitative and quantitative feedback garnered through this study is used to hone the application and in turn, enhance the application experience.
There are minimal risks for you to participate in this study. All personal information will be kept confidential in password-protected computer directories. Your participation will not affect how well you do in your courses (if you are a student) or affect your relationship with the university in any way.
You are free to decline to participate in this study. Should you decide to participate, you are free to end your participation at any time. Such a decision by you will not adversely affect or alter your status with the university in any way.

1. Enter your assigned participant number.

Enter your answer

2. I certify that I have read the preceding and that I understand it's contents. Any questions I have pertaining to the research have been and will be answered by the team. My consent below means that I have freely agreed to participate in this study, and that I agree to the publication of the results for scientific purposes and to the distribution of the recordings and transcripts of the sessions for research purposes so long as my identity is not revealed.

- ◯ I agree
- ◯ I disagree

i. Information collected in this Consent form:

1. Participant Number (To keep participants anonymous)
2. Consent to participate in this usability study for our PowerHouse application.

9.5 Usability Study Questionnaires

i. Pre-testing Questionnaire

[Stage 2] Pre Testing Questionnaire

The objective of this questionnaire is to gather more information about the sample demographics.

1. Please enter your participant number (As assigned in the consent form.) *

Enter your answer

2. Select your gender *

- ◯ Woman
- ◯ Man
- ◯ Prefer not to say
- ◯ Other

3. Select your age group *

- ◯ 18-20
- ◯ 21-25
- ◯ 26-30
- ◯ 31-35
- ◯ 36-40
- ◯ 41-45
- ◯ Above 45

4. State your occupation. *

Enter your answer

5. State your field of occupation/study. *

Enter your answer

6. Do you have prior experience using smart home applications centered around sustainable practices? *

- ◯ Yes
- ◯ No

7. If yes, state them.

Enter your answer

8. How often do you use smart home applications?

- ◯ Daily
- ◯ Weekly
- ◯ Monthly
- ◯ Yearly
- ◯ Never

9. Rate how satisfied you are with the current smart home applications available, on a scale of 1-5. (1 being extremely dissatisfied and 5 being extremely satisfied.)

1	2	3	4	5

10. Which smart home application features would be most important to you? *

- ☐ Remote Device Control
- ☐ Energy Analytics
- ☐ Device Command Automation
- ☐ Energy Saving Tips
- ☐ Grouped Device Control
- ☐ Profile Restrictions (Controlled access)
- ☐ Real-time Energy Breakdown
- ☐ Other

ii. Testing Protocol

[STAGE 2] TESTING PROTOCOL

Section 1

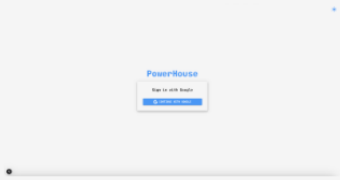
1

Please enter your participant number (As assigned in the consent form.) *

Enter your answer

2

On a scale of 1-5 (1 being extremely difficult and 5 being extremely easy), rate the ease of completing the tasks on this page. [LOGIN PAGE]
Tasks assigned for this page: 1. Login as a user.
2. Add new user and set password. 3. Choose the super-user profile. 4. Unlock profile. 15. Explore sub-user profile. *



1	2	3	4	5

3

Did you find any difficulty in completing the tasks assigned to this page? If yes, please specify.

Enter your answer

4

On a scale of 1-5 (1 being extremely difficult and 5 being extremely easy), rate the ease of completing the tasks on this page. [Main Dashboard]
Tasks assigned for this page: 5. Turn off Philips Hue Smart Bulb B. 6. Investigate daily/monthly energy usage. 12. Switch to light/dark mode. 14. Allocate devices to your newly created profile. *

1	2	3	4	5

5

Did you find any difficulty in completing the tasks assigned to this page? If yes, please specify.

Enter your answer

6

On a scale of 1-5 (1 being extremely difficult and 5 being extremely easy), rate the ease of completing the tasks on this page. [Devices Page]
Tasks assigned for this page: 7. Edit a device name. 8. Delete the Smart Fridge. *

1	2	3	4	5

7

Did you find any difficulty in completing the tasks assigned to this page? If yes, please specify.

Enter your answer

8

On a scale of 1-5 (1 being extremely difficult and 5 being extremely easy), rate the ease of completing the tasks on this page. [Groups Page]
Tasks assigned for this page: 9. Turn on all devices in bedroom. *

1	2	3	4	5

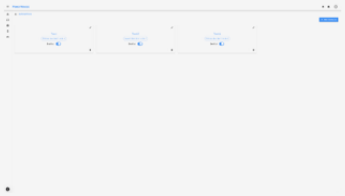
9

Did you find any difficulty in completing the tasks assigned to this page? If yes, please specify.

Enter your answer

10

On a scale of 1-5 (1 being extremely difficult and 5 being extremely easy), rate the ease of completing the tasks on this page. [Automations Page]
Tasks assigned for this page: 10. Add a new automation schedule. *



1	2	3	4	5

11

Did you find any difficulty in completing the tasks assigned to this page? If yes, please specify.

Enter your answer

12

On a scale of 1-5 (1 being extremely difficult and 5 being extremely easy), rate the ease of completing the tasks on this page. [Energy Page]
Tasks assigned for this page: 11. Generate Energy analytics PDF. 13. Set a new energy goal. *

1	2	3	4	5

13

Did you find any difficulty in completing the tasks assigned to this page? If yes, please specify.

Enter your answer

iii. Post-Testing Questionnaire

[Stage 2] Post Testing Questionnaire

The objective of this questionnaire is to collect feedback on the usability of the application, post prototype testing.

1. Please enter your participant number (As assigned in the consent form.) *

Enter your answer

2. On a scale of 1-5, rate your overall user experience with our application. (1 being extremely dissatisfied and 5 being extremely satisfied.) *

1	2	3	4	5

3. On a scale of 1-5, rate the ease in user experience with our application. (1 being extremely dissatisfied and 5 being extremely satisfied.) *

1	2	3	4	5

4. On a scale of 1-5, rate how intuitive the user experience is with our application. (1 being extremely dissatisfied and 5 being extremely satisfied.) *

1	2	3	4	5

5. On a scale of 1-5, rate the appearance of our application (Factors like layout and color). (1 being extremely dissatisfied and 5 being extremely satisfied.) *

1	2	3	4	5

6. On a scale of 1-5, rate how likely you are to recommend our application to others. (1 being extremely unlikely and 5 being extremely likely.) *

1	2	3	4	5

7. Did you manage to complete the specified tasks? *

Enter your answer

8. Please mention what feature of our smart home app you found most useful. *

Enter your answer

9. Please mention what you would like to be improved in terms of usability on our application. *

Enter your answer

10. Please mention what you would like to be improved in terms of design on our application. *

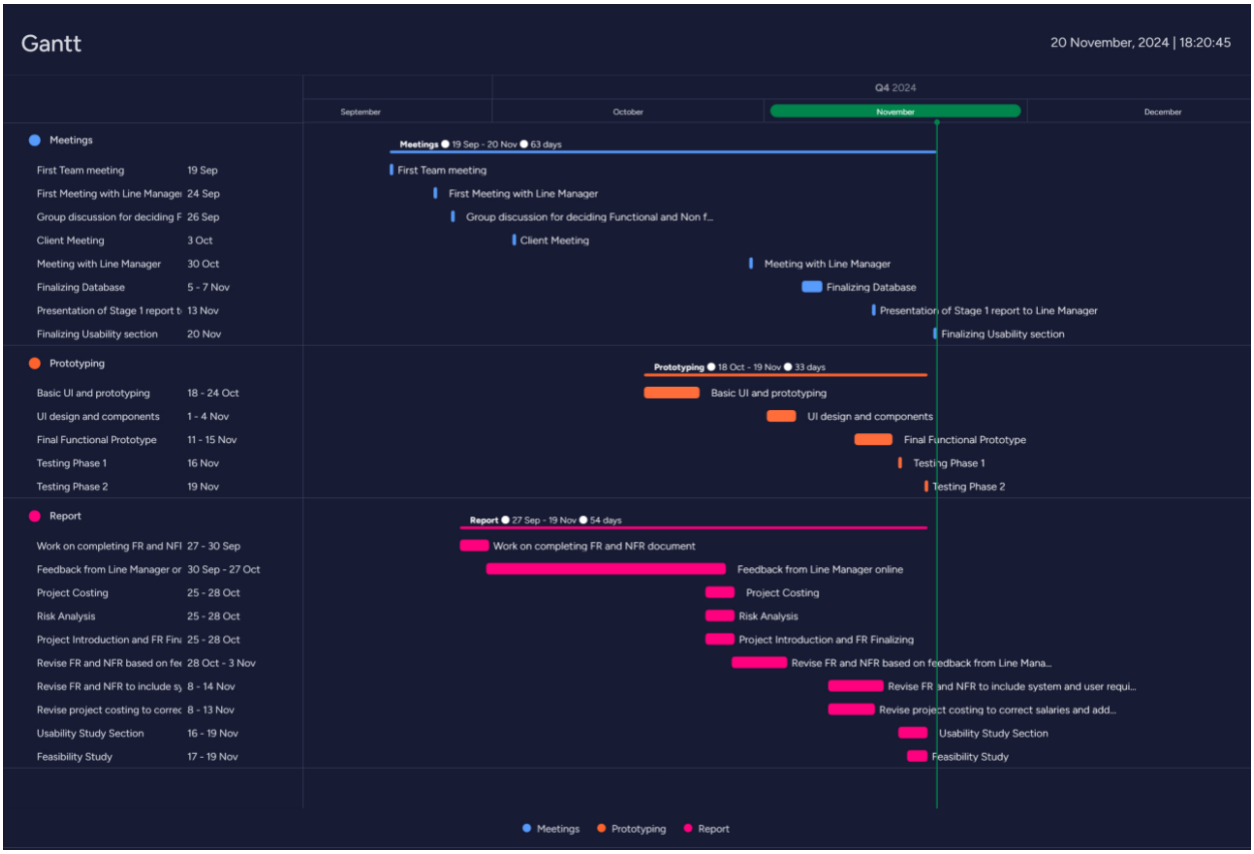
Enter your answer

11. Please mention what you would like to be improved in terms of functionality on our application. *

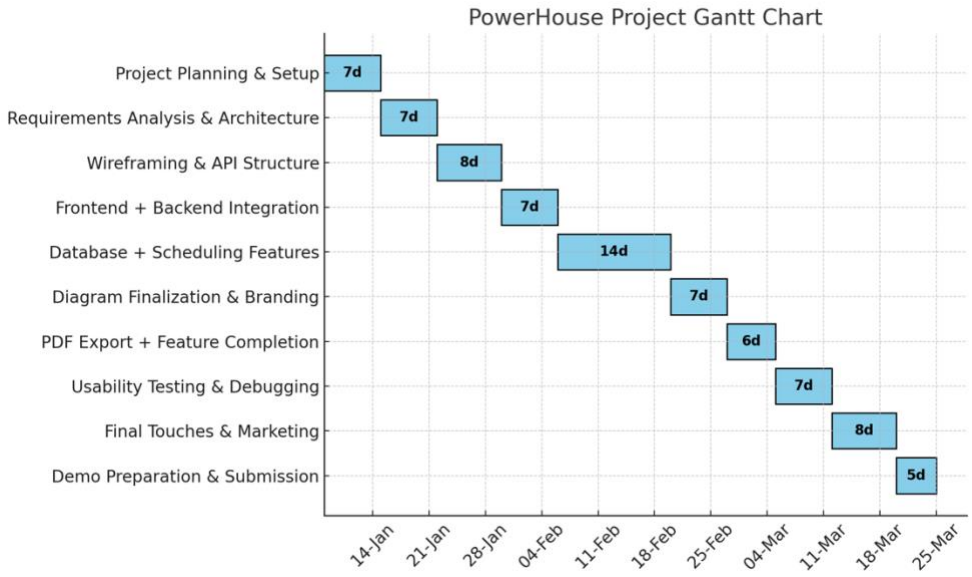
Enter your answer

9.6 Project Schedule

Stage 1

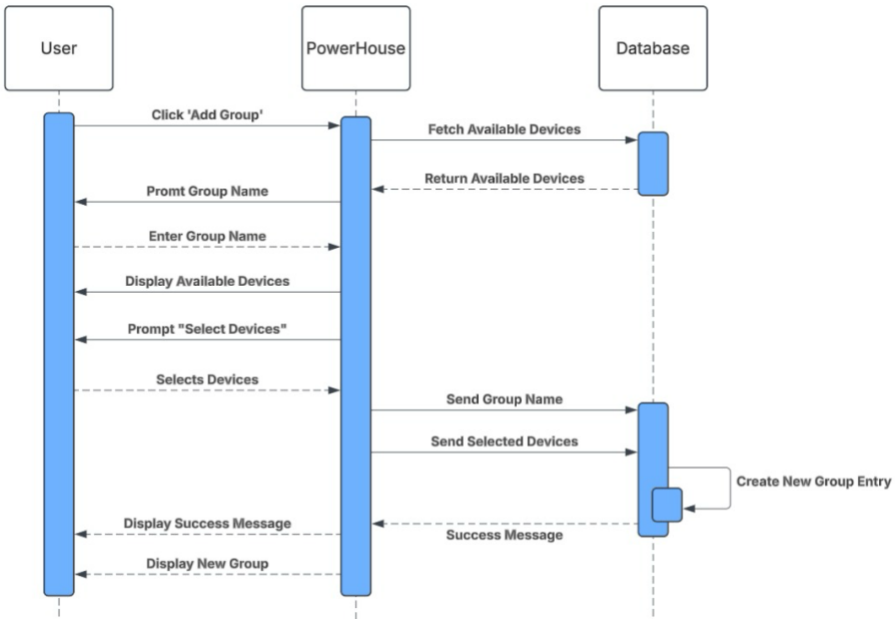


Stage 2

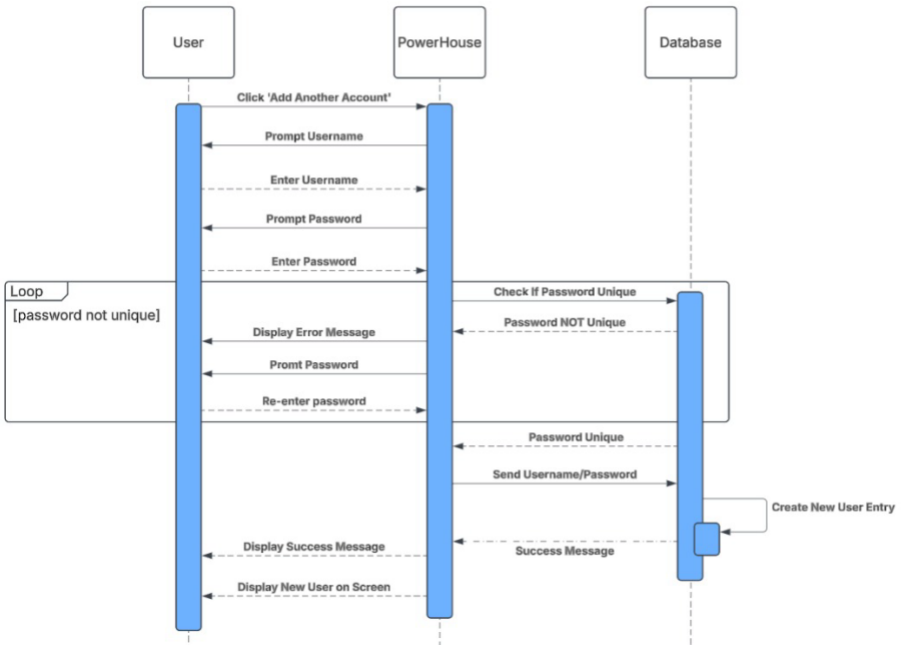


9.7 UML Diagrams

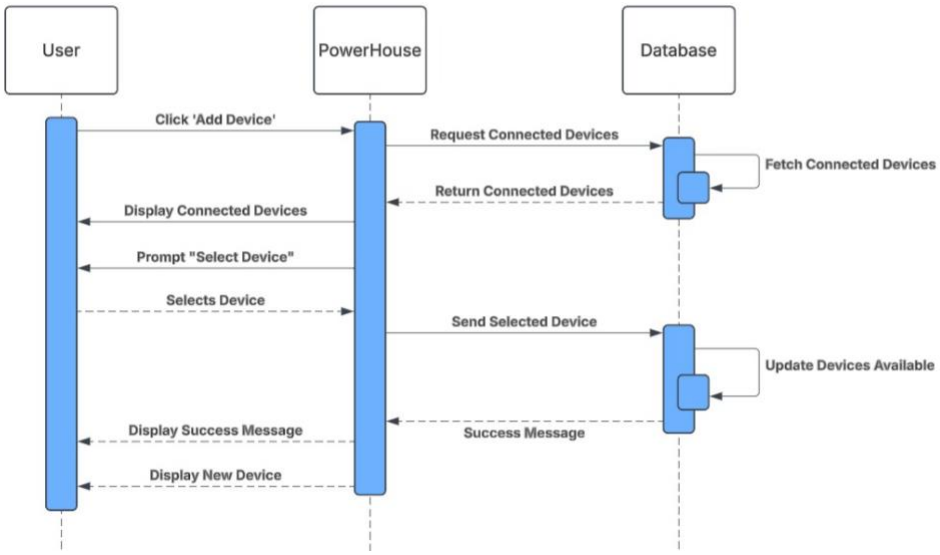
Adding a Device Group



Adding a new user account



Adding a device



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